

FALL 2026 NE 630 HOMEWORK GRADING MANUAL

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This instructor-only manual is built in parallel with the HTML homework assignments. Each problem statement starts on a new page and is followed by its solution for grading and exam-design reference.

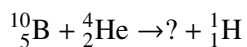
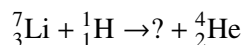
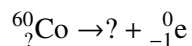
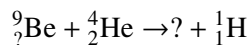
Purple blocks labeled **SOLUTION** reproduce existing course solution language wherever a matching solution exists. Blocks labeled **DRAFT SOLUTION** are new, and blocks labeled **SOLUTION (adapted for Fall 2026)** retain earlier solution language as their foundation but address a changed assignment. Source provenance and the few mathematical or typesetting corrections to legacy material are recorded in comments in the LaTeX source.

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1. HOMEWORK 01: LESSON 01

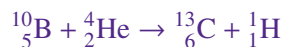
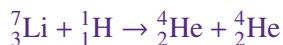
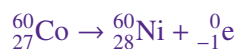
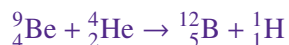
Problem 1 [FNRP 1.3]. Complete the following reactions and compute the corresponding Q values using the mass data from https://www-nds.iaea.org/amdc/ame2020/mass_1.mas20.txt:



When computing Q values, be careful not to round prematurely!

SOLUTION:

The completed reactions are:



The corresponding Q values are (where most units are suppressed for brevity):

$$\begin{aligned} Q_1 &= 931.5 \times ((9.012183062 + 4.00260325413) - (12.014352638 + 1.007825031898)) \\ &\approx -6.885 \text{ MeV} \end{aligned}$$

$$\begin{aligned} Q_2 &= 931.5 \times ((59.933815536) - (59.930785129)) \\ &\approx 2.823 \text{ MeV} \end{aligned}$$

$$\begin{aligned} Q_3 &= 931.5 \times ((7.01600343426 + 1.007825031898) - (4.00260325413 + 4.00260325413)) \\ &\approx 17.346 \text{ MeV} \end{aligned}$$

$$\begin{aligned} Q_4 &= 931.5 \times ((10.012936862 + 4.00260325413) - (13.00335483534 + 1.007825031898)) \\ &\approx 4.062 \text{ MeV} \end{aligned}$$

Problem 2 [FNRP 1.5]. The average kinetic energy of a fission neutron is 2.0 MeV. Defining the kinetic energy as $K = E_{\text{total}} - m_0c^2$, what is the percent error introduced into the kinetic energy from using Eq. (1.12) instead of Eq. (1.9)? (Note that v should be computed using relativistic expressions and that the error should be defined relative to the true value, i.e., $\%err = 100(K_{\text{approximate}} - K_{\text{true}})/K_{\text{true}}$, where the “true” value comes from Eq. 1.9 and the “approximate” value comes from Eq. 1.12.)

SOLUTION:

From Eqs. 1.9 and 1.10, the relativistic expression for total energy is

$$E_{\text{total}} = mc^2 = \frac{m_0c^2}{\sqrt{1 - (v/c)^2}},$$

and, following the problem statement, the relativistic expression for kinetic energy is

$$K_{\text{true}} = mc^2 - m_0c^2 = m_0c^2 \left(\frac{1}{\sqrt{1 - (v/c)^2}} - 1 \right).$$

Equation 1.12 gives the *approximate* expression

$$E_{\text{total}} \approx m_0c^2 + \frac{1}{2}m_0v^2,$$

so that

$$K_{\text{approximate}} = m_0c^2 + \frac{1}{2}m_0v^2 - m_0c^2 = \frac{1}{2}m_0v^2,$$

which is the familiar “classical” expression for kinetic energy.

Given the kinetic energy is 2 MeV, we can use the expression for K_{true} to determine $(v/c)^2$. The rest mass of a neutron is $m_0 = 1.008664$ u. Recall (or remember for next time) that $1 \text{ u} \approx 931.5 \text{ MeV}/c^2$. Thus, we have

$$2 \text{ MeV} = \overbrace{1.008664 \text{ u} \cdot (931.5 \text{ (MeV}/c^2)/\text{u})}^{m_0} \cdot c^2 \left(\frac{1}{\sqrt{1 - (v/c)^2}} - 1 \right)$$

or

$$0.00212821 = \frac{1}{\sqrt{1 - (v/c)^2}} - 1 \rightarrow \boxed{(v/c)^2 \approx 0.0042437}.$$

With $c = 2.99792458 \times 10^8 \text{ m/s}$, we have $v \approx 1.953 \cdot 10^7 \text{ m/s}$. Finally, the error is

$$\begin{aligned} \%err &= 100 \frac{\frac{1}{2}m_0v^2 - 2 \text{ MeV}}{2 \text{ MeV}} \\ &= 100 \frac{\frac{1}{2}1.008664 \text{ u} \cdot (931.5 \text{ (MeV}/c^2)/\text{u})v^2 - 2 \text{ MeV}}{2 \text{ MeV}} \\ &= 100 \frac{469.785258(v^2/c^2) \text{ MeV} - 2 \text{ MeV}}{2 \text{ MeV}} \\ &= 100 \frac{469.785258(0.0042437) - 2 \text{ MeV}}{2 \text{ MeV}} \\ &= \boxed{-0.3184\%}. \end{aligned}$$

Because this error is small, and because the neutron energies of interest to nuclear engineers are typically below 10 MeV, classical expressions are used in practice. As we go through the next few weeks and explore things like cross sections, consider what impact such an approximation may have on the quantities we compute!

Problem 3 [FNRP 1.6]. Consider the following nuclear and chemical reactions:

- A uranium-235 nucleus fissions as a result of being bombarded by a slow neutron. If the energy of fission is 200 MeV, approximately what fraction of the reactants mass is converted to energy?
- A carbon-12 atom undergoes combustion following collision with an oxygen-16 molecule (O_2), forming carbon dioxide. If four eV of energy are released, approximately what fraction of the reactants mass is converted to energy?
- Are the numbers in (a) and (b) consistent with the estimated 20 tons of nuclear fuel per year or the 10,000 tons of coal per day needed for a 1000 MWe nuclear or coal power plant? State your assumptions! (*Hint*: remember from NE 495 that the number of atoms N in a given mass m is given by $N = mN_a/M$, where $N_a = 6.022 \cdot 10^{23}$ /mol is Avogadro's number and M is the atomic mass that we often approximate as the mass number $A = Z + N$ in units of grams per mole. Example: in 100 g of ^{235}U , there are approximately $100(6.022 \cdot 10^{23})/235 \approx 2.56 \cdot 10^{23}$ atoms.)

SOLUTION:

- The total (rest) mass of the reactants is approximately

$$m_0 = 236 \text{ u} \cdot 931.5[\text{MeV}/c^2]/\text{u} \\ \approx 219834.0 \text{ MeV}/c^2.$$

The energy released from fission ΔE is related to the change in mass Δm_0 by $\Delta E = \Delta m_0 c^2$. Thus, $200 \text{ MeV} = \Delta m_0 c^2$, or $\Delta m_0 = 200 \text{ MeV}/c^2$. The relative change in mass is $\Delta m_0/m_0 \approx 200/219834 \approx$ 0.00091 .

- The total (rest) mass of the reactants is approximately

$$m_0 = (12 + 2 \cdot 16) \text{ u} \cdot 931.5[\text{MeV}/c^2]/\text{u} \\ \approx 40986.0 \text{ MeV}/c^2.$$

The energy released from combustion ΔE is related to the change in mass Δm_0 by $\Delta E = \Delta m_0 c^2$. Thus, $4 \cdot 10^{-6} \text{ MeV} = \Delta m_0 c^2$, or

$$\Delta m_0 = 4 \cdot 10^{-6} \text{ MeV}/c^2.$$

The relative change in mass is

$$\frac{\Delta m_0}{m_0} \approx \frac{4 \cdot 10^{-6}}{40986.0} \approx \text{ $9.76 \cdot 10^{-11}$ }.$$

- For a nuclear plant operating at 1000 MWe with 30% efficiency, the total energy produced in one year is

$$\frac{1000 \cdot 10^6}{0.3} \times 365 \cdot 24 \cdot 3600 \approx 1.05 \cdot 10^{17} [\text{J}].$$

The mass of the fuel used is 20 tons, but that mass is largely unchanged during the year! Rather, those 20 tons consist of UO_2 with the uranium enriched to 3-4%. As a very crude estimate, ignore the oxygen, and assume 3% enrichment to find the mass and number of nuclei of uranium 235:

$$N_{U-235} = (20 \cdot 10^6 \times 0.03) \times 6.022 \cdot 10^{23}/235 \approx 1.5 \cdot 10^{27} [\text{nuclei}].$$

If all of those fission, we'd get

$$N_{U-235} \times 200 \times 1.6 \cdot 10^{-13} \approx 4.9 \times 10^{16} [\text{J}].$$

That's within a factor of 2 of the first number, so "sanity" prevails. Beyond the stated assumptions, another reason the first number is larger is that energy is produced not just from U-235; rather, while most of the U-235 is consumed (spent fuel has less U-235 than does natural U), a variety of other fissile nuclides are produced during operations, namely Pu-239, which provides a large fraction of fission power toward the end of the fuel's time in the reactor. We'll cover these "long-term" effects later in the course.

For a coal plant operating at 1000 MWe with 40% efficiency, the total energy produced in one day is

$$\frac{1000 \cdot 10^6}{0.4} \times 24 \cdot 3600 \approx 2.16 \cdot 10^{14} [\text{J}].$$

If we assume coal is 100% carbon, then the 10000 tons of coal consumed in one day involves

$$N_{\text{C-12}} = 10000 \cdot 10^6 \times 6.022 \cdot 10^{23} / 12 \approx 5 \cdot 10^{32} [\text{atoms}] .$$

If all of these combust, then the total energy produced is

$$N_{\text{C-12}} \times 4 \times 1.6 \cdot 10^{-19} \approx 3.2 \cdot 10^{14} [\text{J}] .$$

Again, this is close, so we've passed the "sanity" check.

2. HOMEWORK 02: LESSON 02

Problem 1 [ENRP 1.9]. A reactor operates at a power of 10^3 MW(t) for 1 year. Calculate the power from decay heat

- (a) 1 day following shutdown
- (b) 1 month following shutdown
- (c) 1 year following shutdown
- (d) Repeat a, b, and c, assuming only one month of operation, and compare results.

SOLUTION:

Equation (1.27) provides the Wigner-Way formula for decay heat:

$$P_d(t) = 0.0622P_0(t^{-0.2} - (t_0 + t)^{-0.2}).$$

For all parts, $P_0 = 10^3$ MW. For (a)–(c), $t_0 = 1$ y = 31536000s. Then, for (a), $t = 24 \times 3600 = 86400$ s and

$$P_d(86400) = 0.0622 \times 10^3 \times (86400^{-0.2} - (31536000 + 86400)^{-0.2}) \approx \boxed{4.44 \text{ MW}}$$

For (b), assume that an average month has 30 days so that $t \approx 30 \times 24 \times 3600 = 2592000$ and

$$P_d(2592000) = 0.0622 \times 10^3 \times (2592000^{-0.2} - (31536000 + 2592000)^{-0.2}) \approx \boxed{1.31 \text{ MW}}$$

For (c), $t = t_0$ and

$$P_d(31536000) = 0.0622 \times 10^3 \times (31536000^{-0.2} - (31536000 + 31536000)^{-0.2}) \approx \boxed{0.255 \text{ MW}}$$

For (d), $t_0 = 30 \times 24 \times 3600 = 2592000$ s and

$$P_d(86400) = 0.0622 \times 10^3 \times (86400^{-0.2} - (2592000 + 86400)^{-0.2}) \approx \boxed{3.182 \text{ MW}}$$

$$P_d(2592000) = 0.0622 \times 10^3 \times (2592000^{-0.2} - (2592000 + 2592000)^{-0.2}) \approx \boxed{0.420 \text{ MW}}$$

$$P_d(31536000) = 0.0622 \times 10^3 \times (31536000^{-0.2} - (2592000 + 31536000)^{-0.2}) \approx \boxed{0.031 \text{ MW}}$$

The primary impact of a shorter t_0 is that fewer radioactive fission products are generated and, thus, the decay heat is substantially reduced.

Problem 2 [Lewis Eq. (1.24); adapted from DHNRA 2.19]. Lewis states that more than 40 fission-fragment pairs can emerge from the fission of ^{235}U . Equation (1.24) gives one example, ^{140}Xe and ^{94}Sr . Find one additional representative fission channel published for thermal-neutron-induced fission of ^{235}U . Your source must identify both fragments and the prompt-neutron multiplicity; cite it.

- Write the complete reaction, verify its nucleon-number and charge balance, and compute its Q value using tabulated atomic masses.
- For either Lewis's fragment pair or the pair you selected, compare each fragment's neutron-to-proton ratio with that of a stable isobar having the same mass number.
- Explain why fission fragments are generally neutron rich.
- Predict the principal decay mode of each fragment and explain how that decay moves the products toward stability.
- Identify the particles emitted promptly during fission and those emitted later as the fission products decay. Which contribution to the fission energy is effectively unrecoverable for power production?

SOLUTION (adapted for Fall 2026):

Basic Googling can yield several examples. In addition, there are computer codes that can simulate the fission event using tabulated data. Here, I use a channel identified in the prompt gamma-coincidence measurements of Mukhopadhyay et al. [1]:



The nucleon-number balance is

$$235 + 1 = 96 + 138 + 2 = 236,$$

and the charge balance is

$$92 + 0 = 38 + 54 + 2(0) = 92.$$

Thus, the reaction conserves both nucleon number and charge. Using AME2020 atomic masses [2], the corresponding Q value is

$$\begin{aligned} Q &= 931.494 [(235.043928117 + 1.00866491590) \\ &\quad - (95.921719045 + 137.914146268 + 2(1.00866491590))] \text{ MeV} \\ &\approx \boxed{185.74 \text{ MeV}}. \end{aligned}$$

For the selected products,

$$\left(\frac{N}{Z}\right)_{^{96}\text{Sr}} = \frac{58}{38} \approx 1.526, \quad \left(\frac{N}{Z}\right)_{^{138}\text{Xe}} = \frac{84}{54} \approx 1.556.$$

For comparison, stable isobars ^{96}Mo and ^{138}Ba have

$$\left(\frac{N}{Z}\right)_{^{96}\text{Mo}} = \frac{54}{42} \approx 1.286, \quad \left(\frac{N}{Z}\right)_{^{138}\text{Ba}} = \frac{82}{56} \approx 1.464.$$

Both fission products therefore have larger neutron-to-proton ratios than the corresponding stable isobars.

The compound nucleus $^{236}\text{U}^*$ contains substantially more neutrons than protons. During scission, its nucleons are divided between two fragments on a time scale too short for those fragments to move directly to the valley of stability. Prompt-neutron evaporation removes part of the excess and part of the fragments' excitation energy, but the residual fragments generally remain neutron rich. They subsequently approach stability through beta decay.

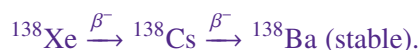
The principal decay mode of both selected fragments is beta-minus decay:

$$n \longrightarrow p + e^- + \bar{\nu}_e.$$

A beta-minus decay leaves A unchanged, increases Z by one, and decreases N by one. It therefore lowers N/Z and moves a neutron-rich isobar toward stability. Representative chains are



and



Excited daughter nuclei can also emit decay gamma rays, and some sufficiently neutron-rich fission products have beta-delayed-neutron branches.

Immediately after fission, the system contains two energetic heavy fragments, prompt neutrons—two in the representative channel above—and prompt gamma rays. At later times, the fission products emit beta particles, electron antineutrinos, decay gamma rays, and, for some precursors, delayed neutrons. The two primary fragments carry roughly 80% of the approximately 200 MeV released in fission. Fragment and beta-particle energy is deposited over short ranges, while neutron and gamma-ray energy is ultimately deposited elsewhere in the reactor and its shielding, apart from leakage. The antineutrinos interact so weakly that their energy is effectively unrecoverable for power production.

Problem 3 [Lewis Eqs. (1.21)–(1.23)]. An idealized prompt-neutron chain begins with n_0 neutrons. Each neutron generation takes an average time ℓ , and the multiplication factor k is constant.

- (a) Beginning with $n_{i+1} = kn_i$, derive

$$n(t) = n_0 k^{t/\ell}.$$

- (b) For $k \approx 1$, show that

$$n(t) \approx n_0 \exp\left(\frac{k-1}{\ell}t\right).$$

- (c) Classify systems having $k = 0.995$, $k = 1.000$, and $k = 1.005$ as subcritical, critical, or supercritical, and describe the long-term behavior of the neutron population in each system.
- (d) A reactor initially produces 1.25 MW(t), with $k = 1.010$ and $\ell = 10^{-5}$ s. Assuming $P(t) \propto n(t)$, estimate the time required to reach 2.00 MW(t) using both the exact and approximate expressions.

DRAFT SOLUTION:

After i neutron generations,

$$n_i = n_0 k^i.$$

If each generation takes an average time ℓ , then $i = t/\ell$. Consequently,

$$n(t) = n_0 k^{t/\ell}.$$

Writing the power as an exponential gives

$$n(t) = n_0 \exp\left(\frac{\ln k}{\ell}t\right).$$

For k near one,

$$\ln k = \ln[1 + (k - 1)] = (k - 1) - \frac{(k - 1)^2}{2} + \dots \approx k - 1.$$

Therefore,

$$n(t) \approx n_0 \exp\left(\frac{k-1}{\ell}t\right).$$

Within this idealized source-free model,

$$k = 0.995 < 1 \implies \text{subcritical; } n(t) \text{ decreases toward zero,}$$

$$k = 1.000 \implies \text{critical; } n(t) = n_0 \text{ remains constant,}$$

$$k = 1.005 > 1 \implies \text{supercritical; } n(t) \text{ increases exponentially.}$$

The unlimited growth in the last line is a consequence of the simplified model; real systems eventually experience feedback and other effects not included here.

Because $P(t) \propto n(t)$,

$$\frac{P(t)}{P(0)} = \frac{2.00}{1.25} = 1.60.$$

Using the exact generation expression,

$$\begin{aligned} t_{\text{exact}} &= \ell \frac{\ln(1.60)}{\ln k} \\ &= (10^{-5} \text{ s}) \frac{\ln(1.60)}{\ln(1.010)} \approx \boxed{4.7235 \times 10^{-4} \text{ s}} = \boxed{0.47235 \text{ ms}}. \end{aligned}$$

Using the $k \approx 1$ approximation,

$$\begin{aligned} t_{\text{approx}} &= \ell \frac{\ln(1.60)}{k-1} \\ &= (10^{-5} \text{ s}) \frac{\ln(1.60)}{0.010} \approx \boxed{4.7000 \times 10^{-4} \text{ s}} = \boxed{0.47000 \text{ ms}}. \end{aligned}$$

The approximation is lower by about 2.35×10^{-6} s, or 0.497%, because $\ln(1.010) < 0.010$.

3. HOMEWORK 03: LESSON 03

Problem 1 [FNRP 1.11]. Uranium-238 has a half-life of 4.51×10^9 yr, whereas the half-life of uranium-235 is only 7.13×10^8 yr. Thus since the earth was formed 4.5 billion years ago, the isotopic abundance of uranium-235 has been steadily decreasing.

- What was the enrichment of uranium when the earth was formed?
- How long ago was the enrichment 4% (by atoms, not mass)?

SOLUTION:

- First, the enrichment is defined as the fraction of the uranium that is ^{235}U . One can define enrichment with respect to either mass or number of atoms; here, we'll use the latter. Further, we **assume** that the only isotopes of uranium are ^{235}U and ^{238}U . Then, at any point in time, the enrichment is

$$e(t) = \frac{N_{235}(t)}{N_{235}(t) + N_{238}(t)} = \frac{\frac{N_{235}(t)}{N_{238}(t)}}{\frac{N_{235}(t)}{N_{238}(t)} + 1} = \frac{R(t)}{R(t) + 1}$$

where, e.g., $N_{235}(t)$ is the number of ^{235}U nuclei on earth at any given time t after the earth was formed, and $R(t) = N_{235}(t)/N_{238}(t)$. Because the number of nuclei of either isotope falls due to radioactive decay, we have

$$N_{235}(t) = N_{235}(0)e^{-\lambda_{235}t} \quad \text{and} \quad N_{238}(t) = N_{238}(0)e^{-\lambda_{238}t}.$$

We don't (and don't need to) know what $N_{235}(0)$ or $N_{238}(0)$ are. It follows that the enrichment is

$$(***) \quad e(t) = \frac{N_{235}(0)e^{-\lambda_{235}t}}{N_{235}(0)e^{-\lambda_{235}t} + N_{238}(0)e^{-\lambda_{238}t}} = \frac{R(0)e^{(\lambda_{238}-\lambda_{235})t}}{R(0)e^{(\lambda_{238}-\lambda_{235})t} + 1}.$$

Today (i.e., $t = 4.5 \cdot 10^9$ y), the isotopic abundance of ^{235}U is 0.72%, so we have

$$0.0072 = \frac{R(0)e^{(\lambda_{238}-\lambda_{235})4.5 \cdot 10^9}}{R(0)e^{(\lambda_{238}-\lambda_{235})4.5 \cdot 10^9} + 1}.$$

From the given half lives and $\lambda = \ln 2/t_{1/2}$, we have

$$\lambda_{238} = 1.5369 \cdot 10^{-10} \text{ y}^{-1} \quad \text{and} \quad \lambda_{235} = 9.7216 \cdot 10^{-10} \text{ y}^{-1}.$$

Then the initial ratio of ^{235}U to ^{238}U is

$$R(0) = \frac{N_{235}(0)}{N_{238}(0)} = \frac{0.0072}{e^{(1.5369-9.7216) \cdot 10^{-10} \cdot 4.5 \cdot 10^9} (1 - 0.0072)} \approx 0.2884.$$

Finally, the initial enrichment is then

$$e(0) = \frac{0.2884}{0.2884 + 1} \approx \boxed{0.2239 = 22.39\%}.$$

- Using (***) with $e(t) = 0.04$ and the now known value for $R(0)$, we have

$$0.04 = \frac{R(0)e^{(\lambda_{238}-\lambda_{235})t}}{R(0)e^{(\lambda_{238}-\lambda_{235})t} + 1},$$

and solving for the unknown time yields

$$t = \frac{\ln\left(\frac{e}{R_0(1-e)}\right)}{\lambda_{238} - \lambda_{235}} = \frac{\ln\left(\frac{0.04}{0.2884(1-0.04)}\right)}{(1.5369 - 9.7216) \cdot 10^{-10}} \approx \boxed{2.36 \cdot 10^9 \text{ y}}.$$

However, because the question asks for "how long ago", we take the current time of 4.5 billion years and subtract t to get $\boxed{2.14 \cdot 10^9 \text{ y}}$.

Problem 2 [FNRP 1.13]. Suppose that a specimen is placed in a reactor, and neutron bombardment causes a radioisotope to be produced at a rate of 2×10^{12} nuclei/s. The radioisotope has a half-life of 2 weeks. How long should the specimen be irradiated to produce 25 Ci of the radioisotope?

SOLUTION:

The appropriate model for this problem is

$$\frac{dN}{dt} = -\lambda N(t) + R, \quad N(0) = 0,$$

the solution of which is

$$N(t) = \frac{R}{\lambda}(1 - e^{-\lambda t}).$$

The corresponding activity is

$$A(t) = \lambda N(t) = R(1 - e^{-\lambda t}),$$

and, for a target activity of A_{target} , we solve for t , or

$$t = -\frac{\log\left(1 - \frac{A_{\text{target}}}{R}\right)}{\lambda}.$$

Substitution of $\lambda = \log 2 / (14 \cdot 24 \cdot 3600) \approx 5.73 \cdot 10^{-7} \text{ s}^{-1}$, $R = 2 \cdot 10^{12} \text{ ns}^{-1}$, and $A_{\text{target}} = 25 \cdot 3.7 \cdot 10^{10} \approx 9.25 \cdot 10^{11} \text{ ns}^{-1}$ into this result yields

$$t = -\frac{\log\left(1 - \frac{9.25 \cdot 10^{11}}{2 \cdot 10^{12}}\right)}{5.73 \cdot 10^{-7}} \approx \boxed{1.083 \cdot 10^6 \text{ s} \approx 12.54 \text{ d}}.$$

Problem 3 [FNRP 1.19]. Consider the fission product chain $A \xrightarrow{\beta} B \xrightarrow{\beta} C$ with decay constants λ_A and λ_B . A reactor is started up at $t = 0$ and produces fission product A at a rate of A_0 thereafter. Assuming that B and C are not produced directly from fission:

- Find $N_A(t)$ and $N_B(t)$.
- What are $N_A(\infty)$ and $N_B(\infty)$?

SOLUTION:

The equations governing the number of each species are

$$\begin{aligned}\frac{dN_A}{dt} &= -\lambda_A N_A(t) + A_0 \\ \frac{dN_B}{dt} &= \lambda_A N_A(t) - \lambda_B N_B(t) \\ \frac{dN_C}{dt} &= \lambda_B N_B(t)\end{aligned}$$

where $N_A(0) = N_B(0) = N_C(0) = 0$. The first equation is decay with constant generation and can be solved using the integrating factor $e^{\lambda_A t}$ as follows:

$$\begin{aligned}\int_0^t \frac{d}{dt'} (N_A(t') e^{\lambda_A t'}) dt' &= \int_0^t A_0 e^{\lambda_A t'} dt' \\ N_A(t) e^{\lambda_A t} - N_A(0) &= \frac{A_0}{\lambda_A} (e^{\lambda_A t} - 1) \\ N_A(t) &= \frac{A_0}{\lambda_A} (1 - e^{-\lambda_A t})\end{aligned}$$

Now the second equation can be solved using the integrating factor $e^{\lambda_B t}$:

$$\begin{aligned}\int_0^t \frac{d}{dt'} (N_B(t') e^{\lambda_B t'}) dt' &= \int_0^t \overbrace{\lambda_A N_A(t')}^{\lambda_A \left(\frac{A_0}{\lambda_A} (1 - e^{-\lambda_A t'}) \right)} e^{\lambda_B t'} dt' \\ N_B(t) e^{\lambda_B t} - N_B(0) &= A_0 \int_0^t (e^{\lambda_B t'} - e^{(\lambda_B - \lambda_A) t'}) dt' \\ N_B(t) e^{\lambda_B t} &= \frac{A_0}{\lambda_B} (e^{\lambda_B t} - 1) - \frac{A_0}{\lambda_B - \lambda_A} (e^{(\lambda_B - \lambda_A) t} - 1) \\ N_B(t) &= \frac{A_0}{\lambda_B} (1 - e^{-\lambda_B t}) - \frac{A_0}{\lambda_B - \lambda_A} (e^{-\lambda_A t} - e^{-\lambda_B t})\end{aligned}$$

It follows that

$$N_A(\infty) = \frac{A_0}{\lambda_A} \quad \text{and} \quad N_B(\infty) = \frac{A_0}{\lambda_B}.$$

Note that for $t = \infty$, the activities of A and B are both equal to A_0 , the rate at which A is produced by the reactor.

4. HOMEWORK 04: LESSON 04

Problem 1 [FNRP 2.2]. The uncollided flux at a distance r from a point source emitting neutrons is given by Eq. (2.9).

- If you are 1 m away from a very small 1-Ci source of neutrons, what is the flux of neutrons in $\text{n/cm}^2/\text{s}$, neglecting scattering and absorption in air?
- If a (1-m thick) shield is placed between you and the source, what absorption cross section would be required to reduce the flux by a factor of 10?
- Suppose the shield made of the material specified in part (b) is only 0.5 m thick. How far must you be from the source for the flux to be reduced by the same amount as in (b)?

SOLUTION:

- Because the source is “very small,” it can be treated as a point source. Without attenuation due to collisions with air, the neutron flux is given by

$$\phi_u = \frac{s_p}{4\pi r^2} = \frac{3.7 \times 10^{10} \text{ n/s}}{4\pi(100)^2 \text{ cm}^2} \approx \boxed{2.94 \cdot 10^5 \frac{\text{n}}{\text{cm}^2 \text{ s}}}.$$

- By placing a 1-m thick shield of material with an absorption cross section equal to Σ_a between you and the source, the flux is attenuated by a factor equal to $e^{-\Sigma_a \cdot 100}$. We want this attenuation to be equal to 0.1, so

$$e^{-\Sigma_a \cdot 100} = 0.1$$

$$\boxed{\Sigma_a = -\frac{\ln(0.1)}{100} \approx 0.023 \frac{1}{\text{cm}}}.$$

- If we have only 50 cm of shielding, the uncollided flux at a distance $r > 50$ from the point source is

$$\phi_u(r) = \frac{s_p e^{-\Sigma_a \cdot 50}}{4\pi r^2}.$$

Set this equal to the flux value of (a) times the attenuation of (b), or

$$\begin{aligned} \frac{s_p e^{-\Sigma_a \cdot 50}}{4\pi r^2} &= \frac{s_p e^{-\Sigma_a \cdot 100}}{4\pi 100^2} \\ \frac{1}{r^2} &= \frac{e^{-\Sigma_a \cdot 50}}{100^2} \\ \frac{1}{r^2} &= \frac{e^{-0.023 \cdot 50}}{100^2} \\ \boxed{r \approx 178 \text{ cm}}. \end{aligned}$$

Problem 2 [FNRP 2.4]. A boiling water reactor operates at 1000 psi. At that pressure, the density of water and of steam are, respectively, 0.74 g/cm^3 and 0.036 g/cm^3 . The microscopic cross sections of H and O are 21.8 b and 3.8 b.

- What is the macroscopic total cross section of the water?
- What is the macroscopic total cross section of the steam?
- If, on average, 40% of the volume is occupied by steam, then what is the macroscopic total cross section of the steam water mixture?
- What is the macroscopic total cross section of water under atmospheric conditions at room temperature?

SOLUTION:

- (a) The number density of water molecules is given by

$$n_w = \frac{\rho_w N_a}{M_w} = \frac{0.74 \cdot 6.022 \cdot 10^{23}}{2 \cdot 1 + 16} \approx 2.4757 \cdot 10^{22} \frac{\text{molecules}}{\text{cm}^3}$$

The number density of oxygen is $n_O = n_w$, while $n_H = 2n_w$. Thus, the macroscopic cross section of water is

$$\begin{aligned} \Sigma_w &= n_O \sigma^O + n_H \sigma^H \\ &= 2.4757 \cdot 10^{22} \frac{\text{a}}{\text{cm}^3} \left(3.8 \frac{\text{b}}{\text{a}} + 2 \cdot 21.8 \frac{\text{b}}{\text{a}} \right) \cdot 10^{-24} \frac{\text{cm}^2}{\text{b}} \\ &\approx \boxed{1.17 \frac{1}{\text{cm}}}. \end{aligned}$$

- (b) Because the number density and, hence, macroscopic cross section scale with the mass density, the macroscopic cross section of steam is

$$\begin{aligned} \Sigma_s &= \frac{\rho_s}{\rho_w} \Sigma_w \\ &= \frac{0.036}{0.74} \cdot 1.17 \\ &\approx \boxed{0.057 \frac{1}{\text{cm}}}. \end{aligned}$$

- (c) Given volume fractions, we can compute an average number density for H and O and recompute a Σ . However, that is equivalent to weighting Σ_w and Σ_s , or

$$\Sigma_{\text{mix}} = 0.4 \Sigma_s + 0.6 \Sigma_w \approx \boxed{0.72 \frac{1}{\text{cm}}}.$$

That the cross section depends so strongly on the steam quality makes accurate descriptions of the water coolant in reactors very important.

- (d) At standard temperature and pressure, $\rho_w \approx 1 \text{ g/cm}^3$, so the modified cross section is

$$\boxed{\Sigma_{w'} = (1/0.74) \Sigma_w = 1.58 \text{ cm}^{-1}}.$$

Problem 3 [FNRP 2.8]. What is the total macroscopic thermal cross section of uranium dioxide (UO_2) that has been enriched to 4%? Assume $\sigma^{235} = 607.5$ b, $\sigma^{238} = 11.8$ b, $\sigma^O = 3.8$ b, and that UO_2 has a density of 10.5 g/cm^3 .

SOLUTION:

First, we assume that the enrichment is by *atomic* composition, which is consistent with the book (see the top of page 36). Then, the molecular mass of 4% enriched UO_2 is

$$M_{\text{UO}_2} = 0.04 \cdot 235 + 0.96 \cdot 238 + 2 \cdot 16 \approx 269.88 \frac{\text{g UO}_2}{\text{mol}},$$

where the mass number of each nuclide is used as the molecular mass (instead of tabulated masses). This introduces a small but negligible error for these sorts of calculations. We can simplify further, too, by assuming $M_U = 238$ (i.e., we assume the same mass for both isotopes) for which $M_{\text{UO}_2} = 270$. I'll keep the first value in my solution.

The number density of UO_2 molecules is given by

$$\begin{aligned} n_{\text{UO}_2} &= \frac{\rho_{\text{UO}_2} N_a}{M_{\text{UO}_2}} \\ &= \frac{10.5 \cdot 6.022 \cdot 10^{23}}{269.88} \\ &\approx 2.3429 \cdot 10^{22} \frac{\text{molecules}}{\text{cm}^3} \end{aligned}$$

Then $n_{235} = 0.04n_{\text{UO}_2}$, $n_{238} = 0.96n_{\text{UO}_2}$, and $n_O = 2n_{\text{UO}_2}$, so that the macroscopic cross section is

$$\begin{aligned} \Sigma_t^{\text{UO}_2} &= n_{235}\sigma_t^{235} + n_{238}\sigma_t^{238} + n_O\sigma_t^O \\ &= n_{\text{UO}_2}(0.04\sigma_t^{235} + 0.96\sigma_t^{238} + 2\sigma_t^O) \\ &= 2.3429 \cdot 10^{22}(0.04(607.5) + 0.96(11.8) + 2(3.8)) \cdot 10^{-24} \\ &\approx \boxed{1.01 \frac{1}{\text{cm}}}. \end{aligned}$$

5. HOMEWORK 05: LESSON 05

Problem 1. Using the BNL NNDC Sigma interface, determine which reactions a neutron can induce, based on the available cross sections, for each of the following nuclides:

- (a) ^1H
- (b) ^{10}B
- (c) ^{238}U

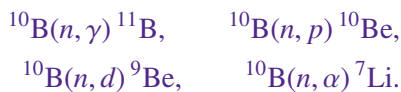
DRAFT SOLUTION:

The available reaction cross sections are:

- (a) For ^1H , the principal reactions are elastic scattering and radiative capture:



- (b) For ^{10}B , the course data include elastic scattering, inelastic scattering, radiative capture, (n, p) , (n, d) , and (n, α) . Representative reactions are



The ^7Li product is often born in an excited state, in which case its subsequent de-excitation also produces a gamma ray.

- (c) For ^{238}U , the principal curves used in the course data are elastic scattering, inelastic scattering, radiative capture, and fission:



together with $^{238}\text{U}(n, n') ^{238}\text{U}^*$ and $^{238}\text{U}(n, f)$. The full evaluated file also includes threshold multiple-neutron reactions such as



once the incident neutron has sufficient energy.

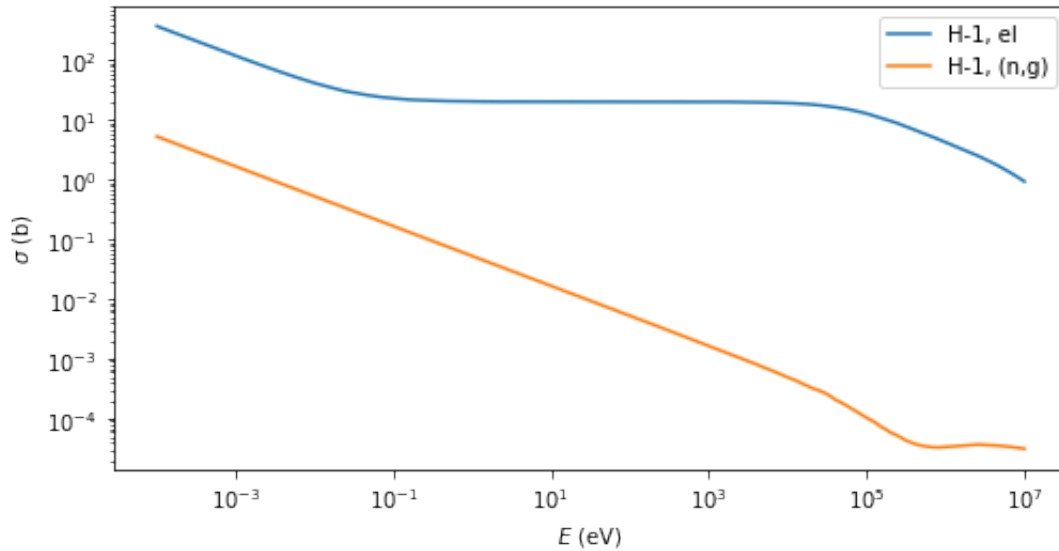
I would not count σ_t as a separate interaction: the total cross section is the sum of the available partial cross sections. Similarly, $\bar{\nu}$ is included in the nuclear data, but it is not a reaction cross section.

Data note: The reaction inventory above follows the archived BNL SIGMA/ENDF data used to prepare the course notebook. The full ENDF file contains more detailed threshold channels than the reduced course data, and the way those channels are grouped can depend on the retrieval selected in SIGMA.

Problem 2. Sketch σ_e and σ_γ for ^1H over $10^{-3} < E < 10^7$ eV. Use log–log axes, with $10^{-4} < \sigma < 10^4$ b on the vertical axis. Here, the subscript e denotes elastic scattering.

DRAFT SOLUTION:

The two cross sections are shown below.



The elastic-scattering cross section is nearly constant at about 20 b over a large part of the low- and intermediate-energy range, and then decreases in the fast range. In contrast, the radiative-capture cross section decreases approximately as $1/v$ over the low-energy range. Because $v \propto \sqrt{E}$,

$$\sigma_\gamma(E) \propto \frac{1}{v} \propto E^{-1/2}.$$

Thus, the capture curve is approximately a straight line with slope $-1/2$ on the log–log plot. The evaluated curve departs from a perfect $1/v$ law at higher energies.

For reference, the values extracted below are

	1 eV	1 MeV
σ_e	20.5933 b	4.28306 b
σ_γ	5.31149×10^{-2} b	3.38313×10^{-5} b.

Data note: The plotted curves and quoted values are from the archived BNL SIGMA/ENDF evaluation used in the Fall 2026 course notebook. A fresh export from another evaluated library or processing temperature may differ slightly.

Problem 3. Using **Evaluated Data** in the Sigma interface, extract the following cross sections at 1 eV and 1 MeV. Use linear interpolation as needed.

Quantity for ^1H	1 eV	1 MeV
σ_e (b)		
σ_γ (b)		
σ_t (b)		

Interpolation note: Given (x_1, y_1) and (x_2, y_2) , evaluate y at x , where $x_1 < x < x_2$, using

$$y(x) = \frac{(x - x_1)y_2 + (x_2 - x)y_1}{x_2 - x_1}.$$

Hint: Based on the available reactions, do you need to look up σ_t , or can you construct it from the partial cross sections?

DRAFT SOLUTION:

Using the interpolation expression in the problem statement, the elastic-scattering cross section at 1 eV is

$$\begin{aligned}\sigma_e(1 \text{ eV}) &= \frac{(1 - 0.40461)(20.2988) + (1.68858 - 1)(20.848)}{1.68858 - 0.40461} \\ &\approx 20.5933 \text{ b.}\end{aligned}$$

Similarly,

$$\begin{aligned}\sigma_\gamma(1 \text{ eV}) &= \frac{(1 - 0.826569)(0.0509196) + (1.07619 - 1)(0.0581121)}{1.07619 - 0.826569} \\ &\approx 0.0531149 \text{ b.}\end{aligned}$$

At 1 MeV, the corresponding interpolations give

$$\begin{aligned}\sigma_e(1 \text{ MeV}) &= \frac{(10^6 - 850000)(4.0527) + (1097760 - 10^6)(4.63652)}{1097760 - 850000} \\ &\approx 4.28306 \text{ b,}\end{aligned}$$

$$\begin{aligned}\sigma_\gamma(1 \text{ MeV}) &= \frac{(10^6 - 800000)(3.67662 \times 10^{-5}) + (2000000 - 10^6)(3.32443 \times 10^{-5})}{2000000 - 800000} \\ &\approx 3.38313 \times 10^{-5} \text{ b.}\end{aligned}$$

At these two energies, the available interactions for ^1H are elastic scattering and radiative capture. Hence,

$$\sigma_t = \sigma_e + \sigma_\gamma.$$

The completed table is

Quantity for ^1H	1 eV	1 MeV
σ_e (b)	20.5933	4.28306
σ_γ (b)	0.0531149	3.38313×10^{-5}
σ_t (b)	20.6464	4.28309

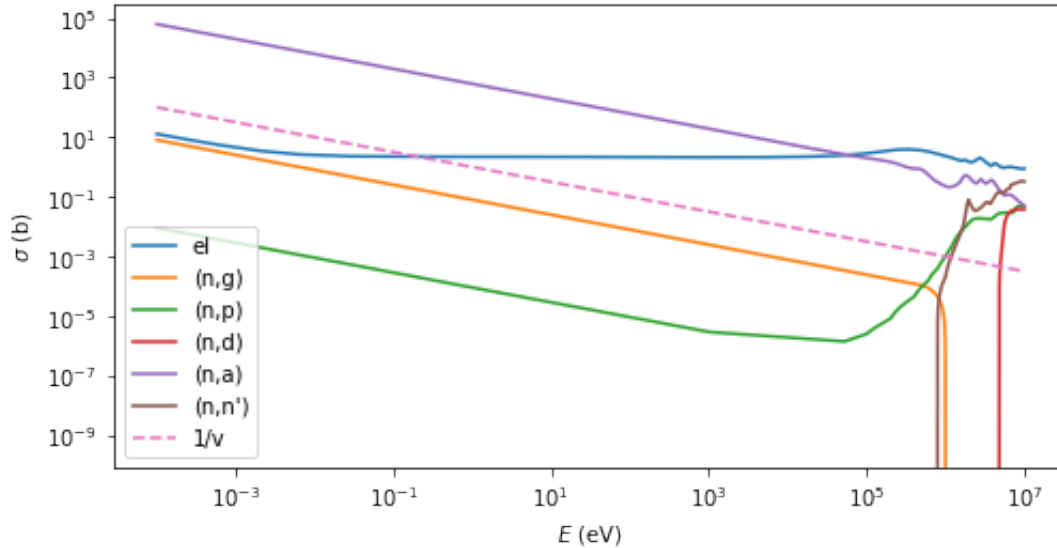
Data note: These numbers use the interpolation points stored in the archived course notebook. Their last quoted digits are evaluation- and interpolation-dependent.

Problem 4. Determine which interaction X dominates the total cross section of ^{10}B . How does $\sigma_X(E)$ depend on E ?

Hint: Straight lines on log–log plots indicate power laws of the form $y(x) = x^p$.

DRAFT SOLUTION:

The available partial cross sections are shown below.



The dominant low-energy interaction is



Away from the higher-energy structure, the (n, α) cross section is approximately a $1/v$ cross section:

$$\sigma_{\alpha}(E) \propto \frac{1}{v} \propto E^{-1/2}.$$

This appears as a straight line with slope $-1/2$ on a log–log plot.

At fast-neutron energies, elastic and inelastic scattering become comparable to, and eventually larger than, the (n, α) cross section. Thus, “dominant” here refers to the broad low-energy range for which ^{10}B is normally used as a neutron absorber; it does not mean that (n, α) is the largest partial cross section at every energy shown.

Problem 5. Using the Sigma interface, determine the missing values in the following table. All cross sections are in barns.

Nuclide	Quantity	1 eV	1 MeV
^{235}U	σ_t		
^{235}U	σ_e		
^{235}U	σ_γ		
^{235}U	σ_f		
^{235}U	$\bar{\nu}$		
^{238}U	σ_t		
^{238}U	σ_e		
^{238}U	σ_γ		
^{238}U	σ_f		
^{238}U	$\bar{\nu}$		

DRAFT SOLUTION:

Here, I'm using the actual text files from BNL Sigma, so these numbers ought to be the same that you get by hand/calculator.

Nuclide	Quantity	1 eV	1 MeV
^{235}U	σ_t (b)	91.60	6.85619
^{235}U	σ_e (b)	13.14	3.65029
^{235}U	σ_γ (b)	10.40	0.108433
^{235}U	σ_f (b)	67.77	1.19979
^{235}U	$\bar{\nu}$	2.421	2.51601
^{238}U	σ_t (b)	9.687	7.12538
^{238}U	σ_e (b)	9.085	4.72403
^{238}U	σ_γ (b)	0.5105	0.128299
^{238}U	σ_f (b)	2.963×10^{-6}	0.0135521
^{238}U	$\bar{\nu}$	2.492	2.56336

The small ^{238}U fission cross section at 1 eV is not a typo. Fission of ^{238}U is possible at low energies in the evaluated data, but it is extremely improbable. Its fission cross section increases substantially in the fast range, which is why ^{238}U is normally described as a threshold-fission nuclide.

Data note: The 1-eV values are retained as rounded constants in the Fall 2026 course notebook; the raw SIGMA text files are not present in the repository. The 1-MeV values are preserved at greater precision in the archived solution notebook. These are values from that archived evaluation, not library-independent constants.

Problem 6. Our TRIGA fuel consists of a mixture of zirconium hydride ($\text{ZrH}_{1.6}$) and uranium metal with a bulk mass density of 7.1 g cm^{-3} .

- Uranium makes up 8.5% of the mixture **by mass**.
- The uranium is enriched to 20% ^{235}U **by atom**; assume the balance is ^{238}U .
- Natural zirconium consists of ^{90}Zr (51.5%), ^{91}Zr (11.2%), ^{92}Zr (17.1%), ^{94}Zr (17.4%), and ^{96}Zr (2.8%), all by atom.
- Assume that ^1H , ^{235}U , and ^{238}U are the only other nuclides.

Use $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$ and approximate each isotope's molar mass by its mass number in g mol^{-1} .

- (a) Compute the indicated molecular and atomic number densities. The following expressions provide a starting point:

$$\overline{M}_{\text{Zr}} = 0.515(90) + 0.112(91) + 0.171(92) + 0.174(94) + 0.028(96),$$

$$M_{\text{ZrH}_{1.6}} = \overline{M}_{\text{Zr}} + 1.6(1), \quad n_{\text{ZrH}_{1.6}} = \frac{(0.915)(7.1)N_A}{M_{\text{ZrH}_{1.6}}},$$

$$n_{^1\text{H}} = 1.6n_{\text{ZrH}_{1.6}}, \quad n_{^{90}\text{Zr}} = 0.515n_{\text{ZrH}_{1.6}},$$

$$\overline{M}_{\text{U}} = 0.20(235) + 0.80(238), \quad n_{\text{U}} = \frac{(0.085)(7.1)N_A}{\overline{M}_{\text{U}}},$$

$$n_{^{235}\text{U}} = 0.20n_{\text{U}}, \quad n_{^{238}\text{U}} = 0.80n_{\text{U}}.$$

Report all six requested densities: $n_{\text{ZrH}_{1.6}}$, $n_{^1\text{H}}$, $n_{^{90}\text{Zr}}$, n_{U} , $n_{^{235}\text{U}}$, and $n_{^{238}\text{U}}$.

- (b) Suppose the elastic-scattering cross section for zirconium, averaged over its naturally occurring isotopes, is 6 b at all neutron energies and that all other interactions with zirconium are negligible. Using the cross sections collected above, compute the macroscopic total, elastic-scattering, and absorption cross sections of the fuel at 1 eV.
- (c) If a 1 eV neutron interacts in the fuel, what is the probability that the interaction is an elastic collision?
- (d) If a 1 eV neutron has an elastic collision, what is the probability that it collides with ^1H ?
- (e) On average, how far does a 1 eV neutron travel in this fuel before undergoing any interaction?
- (f) If a 1 eV neutron is absorbed, what is the probability that it induces fission?
- (g) If a 1 eV neutron induces fission, what is the expected number of emitted neutrons? Account for both uranium isotopes by using a fission-rate-weighted value of $\bar{\nu}$.

DRAFT SOLUTION:

To answer (a)–(g), we need to compute the number density of each nuclide in the fuel mixture and to evaluate macroscopic cross sections for each nuclide and reaction. Given the tabular data and repeated calculations, it is far more convenient to use a short script (in, e.g., Python) or a spreadsheet than to do each calculation by hand. Here, I'm using Python to do the computations.

- (a) First, I'll compute the number densities given the mass density, chemical formula, and uranium enrichment. The average molar mass of natural zirconium is

$$\begin{aligned} \overline{M}_{\text{Zr}} &= 0.515(90) + 0.112(91) + 0.171(92) + 0.174(94) + 0.028(96) \\ &= 91.318 \text{ g/mol}. \end{aligned}$$

Therefore,

$$M_{\text{ZrH}_{1.6}} = 91.318 + 1.6(1) = 92.918 \text{ g/mol}.$$

The zirconium-hydride portion of the fuel has an effective mass density $(1 - 0.085)(7.1) = 6.4965 \text{ g/cm}^3$, and hence

$$\begin{aligned} n_{\text{ZrH}_{1.6}} &= \frac{(0.915)(7.1)(6.022 \times 10^{23})}{92.918} \\ &= 4.21037 \times 10^{22} \text{ cm}^{-3}. \end{aligned}$$

There is one zirconium atom and 1.6 hydrogen atoms per formula unit, so

$$n_{1\text{H}} = 1.6n_{\text{ZrH}_{1.6}} = 6.73659 \times 10^{22} \text{ cm}^{-3},$$

$$n_{90\text{Zr}} = 0.515n_{\text{ZrH}_{1.6}} = 2.16834 \times 10^{22} \text{ cm}^{-3}.$$

For the uranium,

$$\overline{M}_{\text{U}} = 0.20(235) + 0.80(238) = 237.4 \text{ g/mol},$$

so that

$$n_{\text{U}} = \frac{(0.085)(7.1)(6.022 \times 10^{23})}{237.4} = 1.53087 \times 10^{21} \text{ cm}^{-3},$$

$$n_{235\text{U}} = 0.20n_{\text{U}} = 3.06173 \times 10^{20} \text{ cm}^{-3},$$

$$n_{238\text{U}} = 0.80n_{\text{U}} = 1.22469 \times 10^{21} \text{ cm}^{-3}.$$

Thus, all six requested densities are

Quantity	cm^{-3}	$(\text{b cm})^{-1}$
$n_{\text{ZrH}_{1.6}}$	4.21037×10^{22}	0.0421037
$n_{1\text{H}}$	6.73659×10^{22}	0.0673659
$n_{90\text{Zr}}$	2.16834×10^{22}	0.0216834
n_{U}	1.53087×10^{21}	0.00153087
$n_{235\text{U}}$	3.06173×10^{20}	0.000306173
$n_{238\text{U}}$	1.22469×10^{21}	0.00122469

For the individual nuclides, I've adopted the useful unit of a/b-cm. This unit builds in the factor of $10^{-24} \text{ cm}^2/\text{b}$ so that Σ 's can be computed directly.

- (b) Then, I compute the individual nuclide-reaction, macroscopic cross sections. Using the 1-eV values from Problem 5, the hydrogen values from Problem 3, and $\sigma_e^{\text{Zr}} = 6 \text{ b}$ gives

$$\Sigma_e = 10^{-24} [n_{235\text{U}}(13.14) + n_{238\text{U}}(9.085) + n_{\text{ZrH}_{1.6}}(6) + n_{1\text{H}}(20.5933)] \approx \boxed{1.65506 \text{ cm}^{-1}},$$

and

$$\Sigma_a = 10^{-24} [n_{235\text{U}}(10.40 + 67.77) + n_{238\text{U}}(0.5105 + 2.963 \times 10^{-6}) + n_{1\text{H}}(0.0531149)] \approx \boxed{0.0281369 \text{ cm}^{-1}}.$$

Using the tabulated total cross sections,

$$\Sigma_t = 10^{-24} [n_{235\text{U}}(91.60) + n_{238\text{U}}(9.687) + n_{\text{ZrH}_{1.6}}(6) + n_{1\text{H}}(20.6464)] \approx \boxed{1.68340 \text{ cm}^{-1}}.$$

The small difference between Σ_t and $\Sigma_e + \Sigma_a$ is due to the precision retained in the archived, separately interpolated microscopic cross sections and any small channels not listed individually.

- (c) If a neutron interacts, the probability that the interaction is an elastic collision is

$$P(e | t) = \frac{R_e}{R_t} = \frac{\Sigma_e \phi}{\Sigma_t \phi} = \frac{\Sigma_e}{\Sigma_t} = \frac{1.65506}{1.68340} \approx \boxed{0.9832}.$$

- (d) The macroscopic elastic-scattering cross section of hydrogen alone is

$$\Sigma_e^{1\text{H}} = n_{1\text{H}} \sigma_e^{1\text{H}} \approx 1.38729 \text{ cm}^{-1}.$$

Therefore, given that an elastic collision occurred,

$$P(1\text{H} | e) = \frac{\Sigma_e^{1\text{H}}}{\Sigma_e} = \frac{1.38729}{1.65506} \approx \boxed{0.8382}.$$

- (e) The average distance to any interaction is the mean free path:

$$\lambda = \frac{1}{\Sigma_t} = \frac{1}{1.68340} \approx \boxed{0.5940 \text{ cm}}.$$

- (f) The macroscopic fission cross section is

$$\Sigma_f = 10^{-24} [n_{^{235}\text{U}}(67.77) + n_{^{238}\text{U}}(2.963 \times 10^{-6})] = 0.0207494 \text{ cm}^{-1}.$$

Consequently, given that an absorption occurred,

$$P(f | a) = \frac{\Sigma_f}{\Sigma_a} = \frac{0.0207494}{0.0281369} \approx \boxed{0.7374}.$$

- (g) Finally, the fission-rate-weighted neutron yield is

$$\bar{\nu} = \frac{\bar{\nu}_{^{235}\text{U}}\Sigma_f^{^{235}\text{U}} + \bar{\nu}_{^{238}\text{U}}\Sigma_f^{^{238}\text{U}}}{\Sigma_f^{^{235}\text{U}} + \Sigma_f^{^{238}\text{U}}}.$$

Using

$$\Sigma_f^{^{235}\text{U}} = 0.020749364 \text{ cm}^{-1}, \quad \Sigma_f^{^{238}\text{U}} = 3.62877 \times 10^{-9} \text{ cm}^{-1},$$

and $\bar{\nu}_{^{235}\text{U}} = 2.421$ and $\bar{\nu}_{^{238}\text{U}} = 2.492$ gives

$$\boxed{\bar{\nu} \approx 2.421 \text{ neutrons per fission}}.$$

The ^{238}U contribution is negligible at 1 eV, so the result is effectively the ^{235}U value.

Data note: Parts (b)–(g) inherit the rounded, archived BNL SIGMA/ENDF values used in Problems 3 and 5. The displayed arithmetic is internally consistent with those values, but the extra digits should not be interpreted as physical accuracy.

6. HOMEWORK 06: LESSON 06

Problem 1. Use OpenMC's Python nuclear-data interface to examine the radiative-capture cross section, $\sigma_\gamma(E)$, of ^{238}U . No transport calculation is required. On Beocat, the continuous-energy data file is

/homes/jaroberts/openmc/data/neutron/U238.h5.

Load this file with `openmc.data.IncidentNeutron.from_hdf5`, use the radiative-capture reaction (ENDF MT 102), and select the "294K" cross section. The resulting callable takes neutron energy in eV and returns the microscopic cross section in barns.

- Plot $\sigma_\gamma(E)$ over $10^{-4} \leq E \leq 10^7$ eV using log-log axes.
- Above approximately what energy do the individual resonance peaks begin to overlap and appear as a relatively smooth curve?
- Plot $\sigma_\gamma(E)$ again over $1 \leq E \leq 100$ eV using a linear energy axis and a logarithmic cross-section axis.
- At what energy is the first large resonance peak?

Use energy meshes fine enough to resolve the narrow resonances. Increase the mesh resolution until the plotted resonance locations and heights no longer change visibly.

DRAFT SOLUTION:

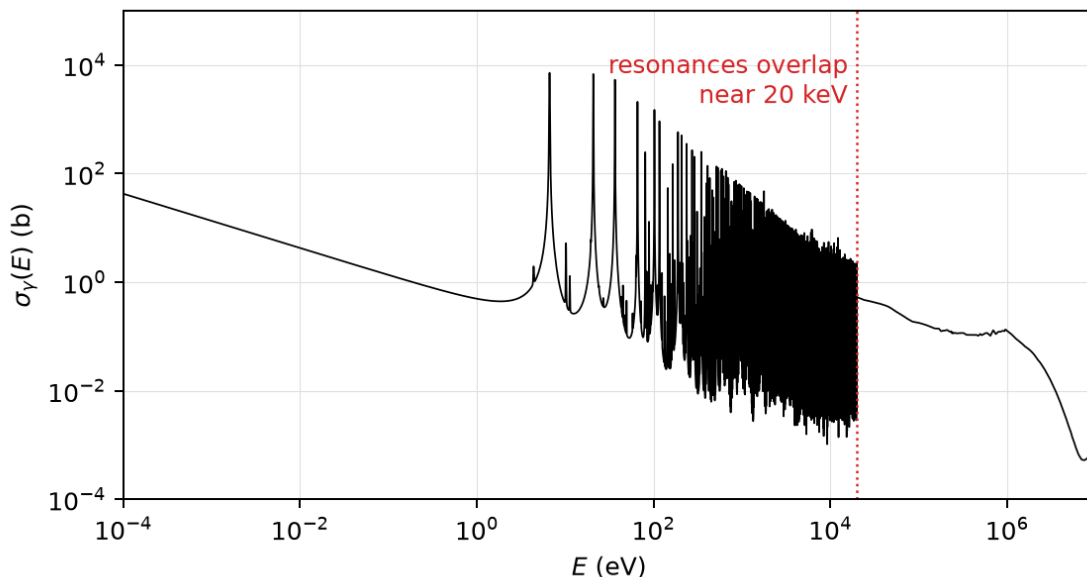
I loaded the 294 K, MT-102 cross section directly from the OpenMC HDF5 file. For example, the data and two plotting meshes can be prepared with

```
import numpy as np
import openmc

data_path = "/homes/jaroberts/openmc/data/neutron/U238.h5"
u238 = openmc.data.IncidentNeutron.from_hdf5(data_path)
sigma_gamma = u238.reactions[102].xs["294K"]

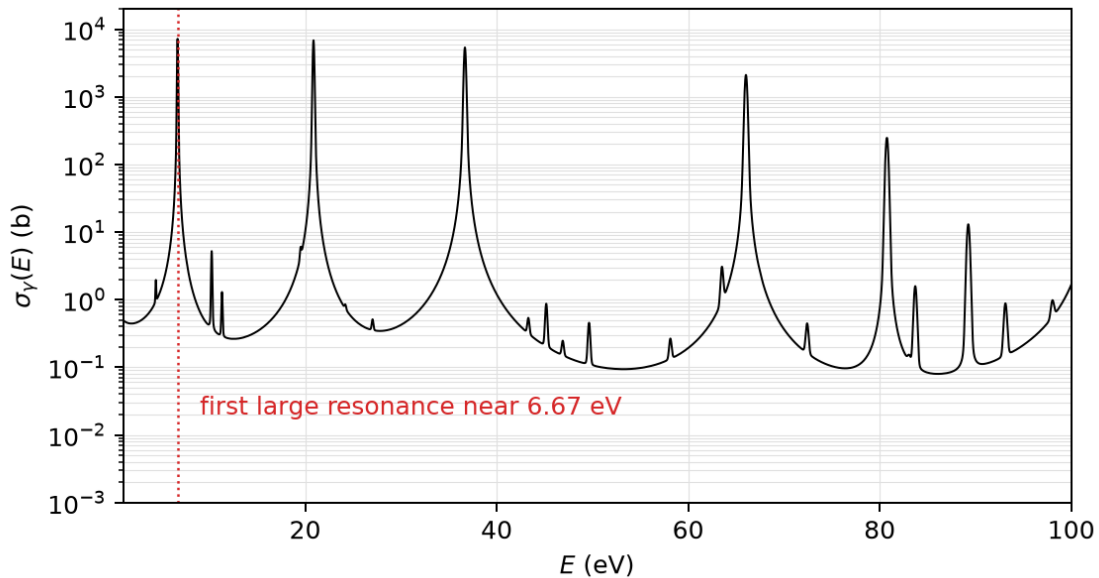
E_full = np.geomspace(1.0e-4, 1.0e7, 500_000)
E_resolved = np.linspace(1.0, 100.0, 1_000_001)
sig_full = sigma_gamma(E_full)
sig_resolved = sigma_gamma(E_resolved)
```

The wide-range result is



At low energy, the resonances are well separated. At about 2×10^4 eV, or 20 keV, they begin to overlap and the evaluated cross section appears relatively smooth on this scale.

Restricting the energy interval gives



The first large resonance is at approximately $E_r = 6.67$ eV. On the native 294 K tabulation used for the figure, the local maximum occurs at 6.67428 eV; the extra digits describe the processed energy grid rather than the accuracy with which the resonance energy should be reported.

I repeated the 1–100 eV plot with successively finer uniform meshes down to $\Delta E \approx 10^{-4}$ eV. The visible peak locations and heights no longer changed at the plotting resolution. Using the native tabulation gives the same plotted result and avoids skipping any of the energy knots stored in the HDF5 file.

Problem 2. Reconstruct the ^{238}U radiative-capture cross section over $1 \leq E \leq 100$ eV with the single-level Breit–Wigner (SLBW) approximation. For resonance i ,

$$\Gamma_i = \Gamma_{n,i} + \Gamma_{\gamma,i}, \quad \lambda_i = \frac{4.55 \times 10^{-10} \text{ cm}}{\sqrt{E_{r,i}/(1 \text{ eV})}},$$

$$\sigma_{0,i} = 4\pi\lambda_i^2 \frac{\Gamma_{n,i}}{\Gamma_i},$$

$$\sigma_{\gamma,i}(E) = \sigma_{0,i} \frac{\Gamma_{\gamma,i}}{\Gamma_i} \left(\frac{E_{r,i}}{E} \right)^{1/2} \frac{1}{1 + 4(E - E_{r,i})^2/\Gamma_i^2},$$

and

$$\sigma_{\gamma}^{\text{SLBW}}(E) = \sum_i \sigma_{\gamma,i}(E).$$

All energies and widths in these expressions are in eV. The expression for $\sigma_{0,i}$ produces an area in cm^2 ; use $1 \text{ b} = 10^{-24} \text{ cm}^2$ to report cross sections in barns. Treat the SLBW reconstruction as a 0 K result.

$E_{r,i}$ (eV)	$\Gamma_{n,i}$ (eV)	$\Gamma_{\gamma,i}$ (eV)
6.67	1.476×10^{-3}	2.300×10^{-2}
20.87	1.009×10^{-2}	2.286×10^{-2}
36.68	3.355×10^{-2}	2.300×10^{-2}
66.03	2.418×10^{-2}	2.331×10^{-2}
80.75	1.874×10^{-3}	2.339×10^{-2}
102.56	7.077×10^{-2}	2.408×10^{-2}

Although the last resonance is centered just above the plotted interval, include its contribution to the sum.

- Compute and plot $\sigma_{\gamma}^{\text{SLBW}}(E)$ over $1 \leq E \leq 100$ eV using a linear energy axis and a logarithmic cross-section axis.
- On the same axes, plot the 294 K OpenMC cross section from Problem 1(c).
- Compare the two curves. Discuss the resonance locations, peak heights, peak widths, and any structure not reproduced by the six-resonance SLBW model. Explain the principal physical reason for the difference in peak height and width between the 0 K SLBW result and the 294 K OpenMC data.

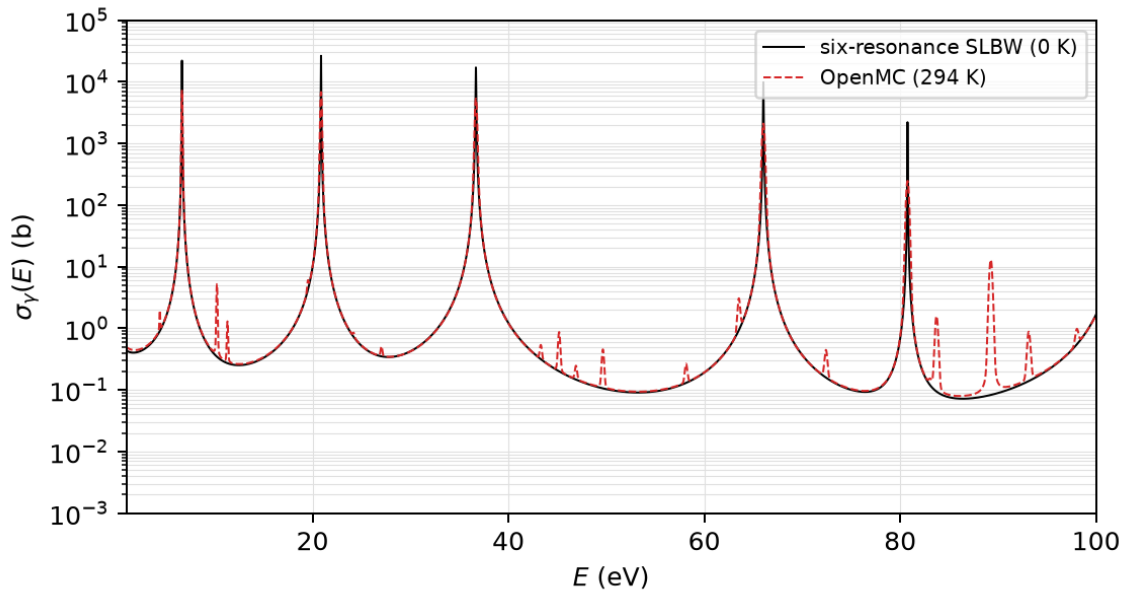
Use a sufficiently fine linear energy grid or local refinement near each resonance, and demonstrate that the reconstructed peaks are converged with respect to the grid resolution.

SOLUTION (adapted for Fall 2026):

The cross-section values reconstructed from the SLBW approximation are shown below with values taken from OpenMC. As in the in-class demo, I've used Python, but Excel or other tools are perfectly acceptable.

```
Er = np.array([6.67, 20.87, 36.68, 66.03, 80.75, 102.56])
Gn = np.array([1.476e-3, 1.009e-2, 3.355e-2,
               2.418e-2, 1.874e-3, 7.077e-2])
Gg = np.array([2.300e-2, 2.286e-2, 2.300e-2,
               2.331e-2, 2.339e-2, 2.408e-2])

E = np.linspace(1.0, 100.0, 1_980_001) # Delta E = 5e-5 eV
sigma_slbw = np.zeros_like(E)
for Er_i, Gn_i, Gg_i in zip(Er, Gn, Gg):
    Gamma_i = Gn_i + Gg_i
    lambda_i = 4.55e-10/np.sqrt(Er_i)
    sigma_0_i = 4*np.pi*lambda_i**2*Gn_i/Gamma_i
    sigma_slbw += (1.0e24*sigma_0_i*(Gg_i/Gamma_i)
                  * np.sqrt(Er_i/E)
                  / (1.0 + 4.0*(E-Er_i)**2/Gamma_i**2))
```



Overall, the SLBW values are quite close in resonance location for the five principal resonances exhibited in the energy range. Representative peak and width values are

E_r (eV)	Γ (eV)	SLBW peak (b)	294 K OpenMC peak (b)
6.67	0.024476	22103	7196
20.87	0.032950	26483	6838
36.68	0.056550	17114	5358
66.03	0.047490	9847	2101
80.75	0.025264	2213	247

For an isolated SLBW resonance, the full width at half maximum is approximately Γ . The corresponding widths read from the 294 K curve are about 0.103, 0.176, 0.240, 0.306, and 0.323 eV, respectively. Thus, the SLBW values exhibit narrower and taller resonances than the OpenMC data.

The six-resonance sum is also missing fine detail and background structure that cannot be captured by this model. Recall from class that the OpenMC data are provided at 294 K. On the other hand, the SLBW model here provides values for 0 K. The increased temperature leads to *Doppler broadening*, which spreads each resonance over a wider energy interval and lowers its peak while approximately preserving its resonance area. The 102.56-eV resonance must be included even though its center lies outside the plot: at 100 eV, its tail contributes 1.669 b of the 1.677 b SLBW total.

I reduced the uniform SLBW grid spacing from 10^{-3} to 5×10^{-5} eV. The resonance locations and quoted peak heights were unchanged at the shown precision, and the plotted curves were visually indistinguishable. The final grid places about 490 points across even the narrowest total resonance width.

Problem 3 [adapted from FNRP 2.9]. The fission-neutron energy spectrum is modeled by

$$\chi(E) = C \exp(-1.036E) \sinh(\sqrt{2.29E}), \quad E \geq 0,$$

where E is in MeV, $\chi(E)$ is a probability density per MeV, and C is an unknown normalization constant. Use numerical integration and the normalized form of $\chi(E)$ to determine the following. Report C and each fraction to at least three significant figures.

- (a) Evaluate the required integral over the whole domain, $0 \leq E < \infty$, and determine C such that

$$\int_0^{\infty} \chi(E) dE = 1.$$

- (b) Evaluate the fraction of fission neutrons born with energies in $0 \leq E \leq 0.1$ MeV.
 (c) Evaluate the fraction of fission neutrons born with energies in $10 \text{ MeV} \leq E < \infty$.

Integrate over each stated interval directly. If the numerical method replaces ∞ with a finite upper limit, demonstrate that the result has converged as that limit is increased.

SOLUTION (adapted for Fall 2026):

The distribution of fission-neutron energies is modeled by

$$\chi(E) = C \exp(-1.036E) \sinh(\sqrt{2.29E}),$$

where E is in MeV. This is definitely a problem best solved using a numerical technique. Here, I'm using `scipy.integrate.quad`:

```
import numpy as np
from scipy.integrate import quad

g = lambda E: np.exp(-1.036*E)*np.sinh(np.sqrt(2.29*E))
I = quad(g, 0.0, np.inf)[0]
C = 1.0/I
chi = lambda E: C*g(E)

p_below = quad(chi, 0.0, 0.1)[0]
p_above = quad(chi, 10.0, np.inf)[0]
```

- (a) The unnormalized integral is

$$I = \int_0^{\infty} \exp(-1.036E) \sinh(\sqrt{2.29E}) dE \approx 2.210125128.$$

Therefore,

$$C = \frac{1}{I} \approx 0.452463070 \text{ MeV}^{-1}.$$

- (b) Direct integration over the requested interval gives

$$\int_0^{0.1} \chi(E) dE \approx \boxed{0.0138794},$$

or about 1.39% of the fission neutrons.

- (c) Direct integration over the high-energy tail gives

$$\int_{10}^{\infty} \chi(E) dE \approx \boxed{0.00106030},$$

or about 0.106% of the fission neutrons.

As a finite-upper-limit check, using upper limits of 20, 30, and 40 MeV for the high-energy-tail integral gives 0.00106008, 0.00106030295, and 0.00106030298, respectively. Likewise, the normalized integral over $0 \leq E \leq 40$ MeV is 0.99999999999996. The reported values are therefore converged well beyond the requested three significant figures.

7. HOMEWORK 07: LESSON 07

Problem 1. Neutrons scatter elastically at 1.0 MeV. After one scattering collision, determine the fraction of the neutrons that will have energies less than 0.5 MeV if they scatter from each of the following:

- (a) Hydrogen
- (b) Deuterium
- (c) Carbon-12
- (d) Uranium-238
- (e) Light water

For part (e), obtain the H-1 and O-16 elastic-scattering cross sections at 1.0 MeV using OpenMC's Python nuclear-data interface (`openmc.data.IncidentNeutron`). Use the elastic-scattering reaction (ENDF MT 2) and the "294K" cross section.

SOLUTION (adapted for Fall 2026):

The probability that a neutron of energy E will scatter down to an energy between E' and $E' + dE'$ after colliding with a target nucleus of mass number A is given by

$$(FNRP 2.47) \quad p(E \rightarrow E') dE' = \begin{cases} \frac{1}{(1-\alpha)E} dE', & \alpha E \leq E' \leq E, \\ 0, & \text{otherwise,} \end{cases}$$

where

$$\alpha = \left(\frac{A-1}{A+1} \right)^2.$$

Then, the probability that a neutron of energy E scatters to an energy equal to or less than E_0 is

$$P(E' \leq E_0) = \begin{cases} \int_{\alpha E}^{E_0} \frac{1}{(1-\alpha)E} dE' = \frac{E_0 - \alpha E}{(1-\alpha)E}, & E_0 > \alpha E, \\ 0, & \text{otherwise.} \end{cases}$$

Thus, for $E = 1$ MeV and $E_0 = 0.5$ MeV, we have for each target nucleus

- (a) $\alpha_H = (1-1)^2/(1+1)^2 = 0$, and

$$P(E' \leq 0.5 \text{ MeV}) = \frac{0.5}{1} = 0.5.$$

- (b) $\alpha_D = (2-1)^2/(2+1)^2 = 1/9$, and

$$P(E' \leq 0.5 \text{ MeV}) = \frac{0.5 - (1/9)1}{(1 - 1/9)1} = 0.4375.$$

- (c) $\alpha_C = (12-1)^2/(12+1)^2 \approx 0.716$ and, hence, $0.5 = E_0 < \alpha_C E = 0.716$, so

$$P(E' \leq 0.5 \text{ MeV}) = 0.$$

- (d) $\alpha_U = (238-1)^2/(238+1)^2 \approx 0.983$, so again

$$P(E' \leq 0.5 \text{ MeV}) = 0.$$

For (e), we have a mixture of two nuclides and, hence, we need

$$(FNRP 2.53) \quad p(E \rightarrow E') = \frac{1}{\Sigma_s(E)} \sum_i \Sigma_{si}(E) p_i(E \rightarrow E').$$

For 1 MeV neutrons, the 294 K OpenMC elastic-scattering cross sections are

$$\sigma_s^{^1\text{H}} = 4.249507 \text{ b}, \quad \sigma_s^{^{16}\text{O}} = 8.125790 \text{ b}.$$

Because $\alpha_{^{16}\text{O}} = (15/17)^2 \approx 0.779$, a 1 MeV neutron cannot scatter off ^{16}O to energies below 0.5 MeV. We already found that half of 1 MeV neutrons scatter off ^1H to energies below 0.5 MeV. Hence, we just need

$$\frac{\Sigma_s^{^1\text{H}}}{\Sigma_s^{\text{water}}} = \frac{2n_w\sigma_s^{^1\text{H}}}{2n_w\sigma_s^{^1\text{H}} + n_w\sigma_s^{^{16}\text{O}}} = \frac{2(4.249507)}{2(4.249507) + 8.125790} \approx 0.511225,$$

where n_w is the number density of water molecules. Therefore,

$$P(E' \leq 0.5 \text{ MeV}) = 0.511225(0.5) \approx 0.255612.$$

Problem 2. Show that, for elastic scattering,

$$\overline{E - E'} \equiv \int_0^E (E - E')p(E \rightarrow E') dE' = \frac{(1 - \alpha)E}{2}.$$

SOLUTION:

Given a probability density function $p(x)$, recall that the expected value of some function $f(x)$ is given by $\bar{f} = \int f(x)p(x) dx$, where the integral is over the domain of x . Here, our density is $p(E \rightarrow E')$, which can also be written as $p(E'; E)$ (i.e., the probability for E' given E as a known parameter). Then, with $f(E') = E - E'$, we have

$$\begin{aligned} \overline{E - E'} &= \int_0^E (E - E')p(E \rightarrow E') dE' \\ &= \int_{\alpha E}^E (E - E') \frac{1}{(1 - \alpha)E} dE' \\ &= \left[\frac{E'}{1 - \alpha} - \frac{E'^2}{2(1 - \alpha)E} \right]_{\alpha E}^E \\ &= \frac{E}{1 - \alpha} \left(\frac{1}{2} - \alpha + \frac{\alpha^2}{2} \right) \\ &= \boxed{\frac{(1 - \alpha)E}{2}}. \end{aligned}$$

8. HOMEWORK 08: LESSON 08

Problem 1. Following the lecture example, use data from OpenMC to reproduce Figure 3.2 from FNRP, and add a curve for U enriched to 4%. This figure should be produced using computer software (e.g., Python, Excel, MATLAB) and *not* by hand.

SOLUTION (adapted for Fall 2026):

A correct solution computes

$$\eta(E) = \frac{e \bar{\nu}_{235}(E) \sigma_{f,235}(E) + (1 - e) \bar{\nu}_{238}(E) \sigma_{f,238}(E)}{e \sigma_{a,235}(E) + (1 - e) \sigma_{a,238}(E)}$$

for the requested enrichments, where e is the atom fraction of ^{235}U . The absorption cross sections should include capture plus fission:

$$\sigma_a(E) = \sigma_\gamma(E) + \sigma_f(E).$$

The plotted curves should reproduce the main behavior of FNRP Fig. 3.2: the natural-uranium curve remains below the enriched cases over most energies, the 20% curve is highest, and the 4% curve lies between them. The raw evaluated data are noisy where the pointwise cross sections and $\bar{\nu}$ are tabulated on different energy grids, so interpolation to a common grid is expected. Smoothing the plotted curve is acceptable for visual comparison, but grading should be based on the unsmoothed values used in any numerical follow-up.

Problem 2. For the 4% values, compute the expected value of $\eta(E)$ using

$$p(E) = \begin{cases} C, & E_{\min} \leq E \leq E_{\max}, \\ 0, & \text{otherwise,} \end{cases}$$

for the following three cases:

- (a) $E_{\min} = 10^{-3}$ eV and $E_{\max} = 1$ eV (the “thermal” range).
- (b) $E_{\min} = 1$ eV and $E_{\max} = 10^5$ eV (the “epithermal” range).
- (c) $E_{\min} = 10^5$ eV and $E_{\max} = 10^7$ eV (the “fast” range).

For each case, define C using the properties of probability density functions; that is,

$$\int_{E_{\min}}^{E_{\max}} p(E) dE = 1.$$

Generally, reactors are classified as being either “thermal” or “fast” spectrum reactors. Based on your values for $\bar{\eta}$ in each energy range, explain why we do not often consider “intermediate” spectrum reactors.

SOLUTION (adapted for Fall 2026):

For a uniform probability density over the interval $[E_{\min}, E_{\max}]$,

$$C = \frac{1}{E_{\max} - E_{\min}}, \quad \bar{\eta} = \frac{1}{E_{\max} - E_{\min}} \int_{E_{\min}}^{E_{\max}} \eta(E) dE.$$

Numerically, the legacy solution notebook evaluates $\eta(E)$ from pointwise BNL/OpenMC-style data and integrates with the trapezoid rule. For the full range $10^{-3} \leq E \leq 10^7$ eV, the same calculation gives

$$\bar{\eta}_{\text{uniform}} \approx 2.82.$$

If instead the flux-weighting density is taken proportional to $1/E$ over the full range, the notebook gives

$$\bar{\eta}_{1/E} = \frac{\int \eta(E) E^{-1} dE}{\int E^{-1} dE} \approx 1.36.$$

Using the local ENDF/B-VIII.1 OpenMC HDF5 data and a converged logarithmic energy grid, the three assigned uniform-range averages are approximately

$$\begin{aligned} 10^{-3} \leq E \leq 1 \text{ eV} : & \quad \bar{\eta} \approx 1.82, \\ 1 \leq E \leq 10^5 \text{ eV} : & \quad \bar{\eta} \approx 0.552, \\ 10^5 \leq E \leq 10^7 \text{ eV} : & \quad \bar{\eta} \approx 2.85. \end{aligned}$$

The epithermal/intermediate range is dominated by strong resonance absorption structure and gives a much lower average reproduction factor than either the thermal or fast range in this calculation. That is the main physical reason reactor spectra are usually pushed toward thermal or fast designs rather than left in an intermediate spectrum.

Problem 3. The slowing-down properties of three common moderators are provided in Table 3.1 of FNRP. Add a row to that table for Be, another element with small mass, and compare Be as a moderator with H₂O, D₂O, and graphite. You may use data from Appendix E, Table E.3.

SOLUTION (adapted for Fall 2026):

Natural Be consists only of ⁹Be and, from Table E.3, has a mass density of $\rho = 1.85 \text{ g/cm}^3$. Thus

$$\alpha = \left(\frac{A-1}{A+1} \right)^2 = \left(\frac{8}{10} \right)^2 = 0.64,$$

and

$$\xi = 1 + \frac{\alpha}{1-\alpha} \ln \alpha \approx 0.207.$$

The number density of natural Be is

$$n_{\text{Be}} = \frac{1.85(6.022 \times 10^{23})}{9.013} \approx 1.24 \times 10^{23} \text{ atoms/cm}^3.$$

Using $\sigma_s = 6.151 \text{ b}$ and $\sigma_a = 0.0085 \text{ b}$,

$$\Sigma_s \approx 0.76 \text{ cm}^{-1}, \quad \Sigma_a \approx 1.1 \times 10^{-3} \text{ cm}^{-1}.$$

Therefore,

$$\xi \Sigma_s \approx 0.157 \text{ cm}^{-1}, \quad \frac{\xi \Sigma_s}{\Sigma_a} \approx 1.5 \times 10^2.$$

Be has a slowing-down decrement and slowing-down power between heavy water and graphite, as expected from its mass. Its slowing-down ratio is better than light water but below graphite. It is plausible as a reflector or specialty moderating material, but graphite and heavy water remain the more familiar natural-uranium moderator choices.

9. HOMEWORK 09: LESSON 09

Problem 1. Verify Eqs. (3.23) and (3.25):

(FNRP 3.23)
$$\Sigma_s(E)\varphi(E) = \frac{C}{E}$$

and

(FNRP 3.25)
$$q = \left[1 + \frac{\alpha}{1-\alpha} \ln \alpha \right] C.$$

Hint: To verify Eq. (3.23), substitute Eq. (3.23) into Eq. (3.22) and show that the left- and right-hand sides are equal.

SOLUTION (adapted for Fall 2026):

Substitute $\Sigma_s(E)\phi(E) = C/E$ into Eq. (3.22):

$$\begin{aligned} \Sigma_s(E)\phi(E) &= \int_E^{E/\alpha} \frac{1}{(1-\alpha)E'} \Sigma_s(E')\phi(E') dE' \\ &= \int_E^{E/\alpha} \frac{1}{(1-\alpha)E'} \frac{C}{E'} dE' \\ &= \frac{C}{1-\alpha} \int_E^{E/\alpha} \frac{dE'}{E'^2} \\ &= \frac{C}{1-\alpha} \left[-\frac{1}{E'} \right]_E^{E/\alpha} \\ &= \frac{C}{1-\alpha} \left(\frac{1}{E} - \frac{\alpha}{E} \right) = \boxed{\frac{C}{E}}. \end{aligned}$$

For Eq. (3.25),

$$\begin{aligned} q &= \int_E^{E/\alpha} \left[\int_{\alpha E'}^E \frac{1}{(1-\alpha)E'} \Sigma_s(E')\phi(E') dE'' \right] dE' \\ &= \frac{C}{1-\alpha} \int_E^{E/\alpha} \frac{E - \alpha E'}{E'^2} dE' \\ &= \frac{C}{1-\alpha} \left(\int_E^{E/\alpha} \frac{E}{E'^2} dE' - \int_E^{E/\alpha} \frac{\alpha}{E'} dE' \right) \\ &= \frac{C}{1-\alpha} (1 - \alpha + \alpha \ln \alpha) \\ &= \boxed{\left[1 + \frac{\alpha}{1-\alpha} \ln \alpha \right] C}. \end{aligned}$$

Problem 2. For thermal neutrons, calculate $\bar{\eta}$ as a function of uranium enrichment and plot your results for enrichments between 0% and 30%. Use the uranium data from the following table:

	ν	σ_f (barns)	σ_a (barns)
Uranium-235	2.43	505	591
Plutonium-239	2.90	698	973
Uranium-238	—	0	2.42

Consider the value you compute for 4% enrichment. How does this value compare with the values computed for Problem 2 of HW08?

SOLUTION (adapted for Fall 2026):

For enrichment e , the thermal reproduction factor for the uranium mixture is

$$\eta(e) = \frac{e \nu_{235} \sigma_{f,235}}{e \sigma_{a,235} + (1 - e) \sigma_{a,238}}.$$

Substitution gives

$$\eta(e) = \frac{e(2.43)(505)}{e(591) + (1 - e)(2.42)}.$$

Thus, for $e = 0.04$,

$$\eta(0.04) = \frac{0.04(2.43)(505)}{0.04(591) + 0.96(2.42)} \approx \boxed{1.89}.$$

A Python plotting solution is

```
import numpy as np
import matplotlib.pyplot as plt

nu_235 = 2.43
sigma_f_235 = 505.0
sigma_a_235 = 591.0
sigma_a_238 = 2.42

enrichment = np.linspace(0, 0.3, 100)
eta = lambda e: e*nu_235*sigma_f_235/(e*sigma_a_235
                                     + (1-e)*sigma_a_238)

plt.plot(100*enrichment, eta(enrichment), "k")
plt.plot(4, eta(0.04), "rx")
plt.xlabel("enrichment (%)")
plt.ylabel(r"$\eta$")
```

The 4% value is close to the thermal-range values from HW08 and to the low energy portion of the 4% $\eta(E)$ curve. It is much more favorable than the resonance-dominated epithermal behavior.

Problem 3. Lethargy, defined as $u = \ln(E_0/E)$, is often used in neutron slowing-down problems; lethargy increases as energy decreases. Note the following transformations:

$$\varphi(E) dE = -\varphi(u) du, \quad p(E \rightarrow E') dE' = -p(u \rightarrow u') du', \quad \Sigma_x(E) = \Sigma_x(u).$$

(a) Show that $p(E \rightarrow E')$, given by Eq. (2.47), becomes

$$p(u \rightarrow u') = \begin{cases} \frac{1}{1-\alpha} \exp(u-u'), & u \leq u' \leq u + \ln(1/\alpha), \\ 0, & \text{otherwise.} \end{cases}$$

(b) Express Eq. (3.22) in terms of u .

(c) Show that the slowing-down decrement can be expressed as $\xi = \overline{\Delta u}$.

SOLUTION (adapted for Fall 2026):

(a) Since $E' = E_0 e^{-u'}$, $dE'/du' = -E'$. Therefore,

$$p(u \rightarrow u') = -p(E \rightarrow E') \frac{dE'}{du'} = \frac{1}{(1-\alpha)E} E' = \frac{1}{1-\alpha} \exp(u-u').$$

The original support $\alpha E \leq E' \leq E$ becomes $u \leq u' \leq u + \ln(1/\alpha)$.

(b) Using $\phi(u) = \phi(E)E$ and $E = E_0 e^{-u}$, Eq. (3.22) becomes

$$\Sigma_s(u)\phi(u) = \int_{u-\ln(1/\alpha)}^u \frac{e^{u'-u}}{1-\alpha} \Sigma_s(u')\phi(u') du'.$$

(c) With the density found in part (a),

$$\begin{aligned} \overline{\Delta u} &= \int_u^{u+\ln(1/\alpha)} \frac{1}{1-\alpha} e^{u-u'} (u' - u) du' \\ &= 1 + \frac{\alpha}{1-\alpha} \ln \alpha \\ &= \boxed{\xi}. \end{aligned}$$

10. HOMEWORK 10: LESSON 10

Problem 1. Use the narrow-resonance (NR) approximation to compare the epithermal flux spectra of homogeneous mixtures of ^1H and ^{238}U . Consider hydrogen-to-uranium atom ratios

$$R = \frac{N_{^1\text{H}}}{N_{^{238}\text{U}}} = 1000, 100, \text{ and } 10.$$

Use OpenMC's Python nuclear-data interface to obtain the ^1H and ^{238}U total microscopic cross sections (ENDF MT 1) at 294 K. Follow the `openmc.data.DataLibrary` and `openmc.data.IncidentNeutron` workflow used in the Lesson 10 notebook. No transport calculation is required for this problem.

For a mixture with atom ratio R , the total macroscopic cross section is proportional to

$$\Sigma_{t,R}(E) \propto R\sigma_t^{^1\text{H}}(E) + \sigma_t^{^{238}\text{U}}(E).$$

The NR spectrum has the shape

$$\tilde{\phi}_R(E) = \frac{1}{\Sigma_{t,R}(E)E}.$$

Normalize each spectrum with

$$\phi_R(E) = \frac{\tilde{\phi}_R(E)}{\tilde{\phi}_R(1 \text{ eV})}, \quad \phi_R(1 \text{ eV}) = 1.$$

- Plot the three normalized spectra over $1 \leq E \leq 100 \text{ eV}$.
- Plot $E\phi_R(E)$ over the same interval.
- Explain how the moderator-to-fuel ratio affects resonance flux depressions and energy self-shielding.
- Without an additional numerical integration, rank the three mixtures by their expected ^{238}U capture rate per ^{238}U nucleus,

$$\frac{R_{\gamma,R}}{N_{^{238}\text{U}}} = \int_{1 \text{ eV}}^{100 \text{ eV}} \sigma_{\gamma}^{^{238}\text{U}}(E)\phi_R(E) dE,$$

under the same normalization. Neglect absorption in hydrogen.

SOLUTION (adapted for Fall 2026):

The calculation can be done directly from the pointwise total cross sections:

```
import numpy as np
import matplotlib.pyplot as plt

# h1 and u238 are openmc.data.IncidentNeutron objects at 294 K.
sig_t_h1 = h1[1].xs["294K"] # ENDF MT 1 total
sig_t_u238 = u238[1].xs["294K"] # ENDF MT 1 total

E = np.linspace(1.0, 100.0, 4000)
for R in (1000, 100, 10):
    phi_tilde = 1.0 / ((R*sig_t_h1(E) + sig_t_u238(E))*E)
    phi = phi_tilde / phi_tilde[0]
    plt.semilogy(E, phi, label=f"{R}-to-1")
plt.xlabel("E [eV]")
plt.ylabel(r"$\phi_R(E)$")
plt.legend()
```

The spectra should show the deepest depressions near ^{238}U resonances for the smallest moderator-to-fuel ratio. Increasing R increases the smooth hydrogen contribution to Σ_t , so the resonant ^{238}U structure is diluted and the flux dips become shallower.

With all spectra normalized to $\phi_R(1 \text{ eV}) = 1$, the expected capture rate per ^{238}U nucleus is largest for the least self-shielded case and smallest for the most self-shielded case:

$$R = 1000 > R = 100 > R = 10.$$

The ranking follows because $\sigma_{\gamma}^{238\text{U}}(E)$ is largest at the same resonance energies where the low- R spectra have the strongest flux depressions.

Problem 2. Adapted from FNRP 3.6.

Consider an infinite, homogeneous mixture containing a moderator m and a resonant fuel f . In the epithermal range, the slowing-down balance is

$$\begin{aligned}\Sigma_t(E)\phi(E) &= \int_E^{E/\alpha_f} \frac{\Sigma_s^f(E')\phi(E')}{(1-\alpha_f)E'} dE' \\ &\quad + \int_E^{E/\alpha_m} \frac{\Sigma_s^m(E')\phi(E')}{(1-\alpha_m)E'} dE' \\ &= I_f(E) + I_m(E).\end{aligned}$$

In the wide-resonance (WR) approximation, also called the narrow-resonance infinite-mass approximation, take $A_f \rightarrow \infty$ so that $\alpha_f \rightarrow 1$. Retain the NR assumptions for the moderator contribution: the moderator scattering cross section is smooth, and the surrounding flux has the background shape C/E .

- Evaluate $I_m(E)$ under the moderator NR assumptions.
- Show that $\lim_{\alpha_f \rightarrow 1} I_f(E) = \Sigma_s^f(E)\phi(E)$.
- Substitute the two scattering-in contributions into the balance equation and solve for $\phi_{\text{WR}}(E)$.
- Compare your WR spectrum with the NR result

$$\phi_{\text{NR}}(E) \approx \frac{C(\Sigma_{s,0}^m + \Sigma_{s,0}^f)}{\Sigma_t(E)E}.$$

- Explain the comparison physically.

SOLUTION (adapted for Fall 2026):

Under the NR assumptions for the moderator,

$$I_m(E) = \int_E^{E/\alpha_m} \frac{\Sigma_s^m}{(1-\alpha_m)E'} \frac{C}{E'} dE' = \frac{C\Sigma_s^m}{E}.$$

For the fuel contribution,

$$\begin{aligned}I_f(E) &= \int_E^{E/\alpha_f} \frac{\Sigma_s^f(E')\phi(E')}{(1-\alpha_f)E'} dE' \\ &\approx \frac{\Sigma_s^f(E)\phi(E)}{(1-\alpha_f)E} \left(\frac{E}{\alpha_f} - E \right) \xrightarrow{\alpha_f \rightarrow 1} \Sigma_s^f(E)\phi(E).\end{aligned}$$

The balance equation becomes

$$\Sigma_t(E)\phi(E) = \frac{C\Sigma_s^m}{E} + \Sigma_s^f(E)\phi(E),$$

so

$$\boxed{\phi_{\text{WR}}(E) = \frac{C\Sigma_s^m}{[\Sigma_t(E) - \Sigma_s^f(E)]E}}.$$

The WR approximation predicts a deeper flux depression through a resonance than the NR approximation. In the WR limit, scattering from the fuel changes the neutron energy negligibly, so it does not provide a broad scattering-in source across the resonance. That leaves the moderator as the only broad source into the resonance energy and increases energy self-shielding.

11. HOMEWORK 11: LESSON 11

Problem 1 [FNRP 3.9]. Suppose that the Maxwell-Boltzmann distribution, Eq. (2.34), represents the neutron density in Eqs. (3.43) and (3.44).

- Find the value of $\bar{v}$.
- If we define $\bar{E} = \frac{1}{2}m\bar{v}^2$, show that $\bar{E} \approx 1.273kT$.
- Why is your result different from the average energy of $\frac{3}{2}kT$ given by Eq. (2.33)?

SOLUTION (adapted for Fall 2026):

The Maxwell-Boltzmann energy distribution is

$$M(E) = \frac{2\pi}{(\pi kT)^{3/2}} E^{1/2} e^{-E/kT}.$$

- (a) Substitution into Eq. (3.44) gives

$$\begin{aligned} \bar{v} &= \frac{\int_0^\infty v(E)M(E) dE}{\int_0^\infty M(E) dE} \\ &= \frac{2\pi}{(\pi kT)^{3/2}} \sqrt{\frac{2}{m}} \int_0^\infty E e^{-E/kT} dE \\ &= \boxed{\frac{2\sqrt{2kT}}{\sqrt{\pi m}}}. \end{aligned}$$

- (b) Therefore,

$$\bar{E} = \frac{1}{2}m\bar{v}^2 = \frac{1}{2}m \left(\frac{2\sqrt{2kT}}{\sqrt{\pi m}} \right)^2 = \boxed{\frac{4kT}{\pi} \approx 1.273kT}.$$

- (c) The results differ because $\bar{E}$ here is computed from the square of the mean speed, while the true expected energy depends on the mean of the speed squared. In general, $(\bar{v})^2 \neq \overline{v^2}$.

Problem 2. Construct a model in OpenMC for an infinite, homogeneous system composed of a 1000-to-1 atom mixture of ^1H and ^{238}U , with a point source of 1 MeV neutrons. Compute the neutron flux spectrum over $10^{-3} < E < 10^5$ eV. Then produce a plot containing:

- The computed flux spectrum.
- The Maxwell-Boltzmann flux spectrum for the selected temperature, up to 1 eV.
- The $1/E$ spectrum above 1 eV.

Here the thermal comparison is a *flux* spectrum, $\phi_{\text{MB}}(E) \propto E \exp[-E/(kT)]$, rather than the neutron-density distribution used in Problem 1. The comparison curves may be scaled to compare their shapes with the computed spectrum; no absolute source strength is specified.

SOLUTION (adapted for Fall 2026):

One representative OpenMC model is

```
import openmc
import numpy as np
import matplotlib.pyplot as plt

model = openmc.Model()
mat = openmc.Material(name="u238_h1")
mat.add_nuclide("H1", 1000.0)
mat.add_nuclide("U238", 1.0)
mat.set_density("g/cm3", 1.0)
mat.temperature = 294
model.materials = openmc.Materials([mat])

sphere = openmc.Sphere(r=1.0, boundary_type="reflective")
cell = openmc.Cell(region=-sphere, fill=mat)
model.geometry = openmc.Geometry([cell])

settings = openmc.Settings()
settings.run_mode = "fixed source"
settings.source = openmc.IndependentSource(
    space=openmc.stats.Point((0, 0, 0)),
    energy=openmc.stats.Discrete([1e6], [1.0]),
)
settings.seed = 1234
settings.batches = 100
settings.particles = 10000
settings.cutoff = {"energy_neutron": 1e-3}
model.settings = settings

energies = np.logspace(-3, 5, 1000)
tally = openmc.Tally()
tally.filters = [openmc.EnergyFilter(energies)]
tally.scores = ["flux"]
model.tallies = [tally]
```

After running the model, plot the tally per unit energy and compare it with scaled shapes:

```
E = energies[:-1]
plt.step(E, flux_mean/np.diff(energies), where="post",
        color="k", lw=0.9, label="simulation")
plt.plot(E[E > 1], const_1/E[E > 1], "r--", label="1/E")
plt.plot(E[E < 1], const_2*E[E < 1]*np.exp(-E[E < 1]/0.0253),
        "b:", label="Maxwell-Boltzmann flux")
plt.xscale("log")
```

```
plt.yscale("log")
plt.xlabel("E [eV]")
plt.ylabel("flux per unit energy")
plt.legend()
```

The simulated spectrum should approach a Maxwell-Boltzmann-like thermal shape below about 1 eV and a slowing-down shape close to $1/E$ above the thermal range, with resonance effects where ^{238}U data are important.

12. HOMEWORK 12: LESSON 12

Problem 1. Equations (3.50)–(3.53) in FNRP describe how to compute the resonance integral. Use these equations with nuclear data from OpenMC to compute the absorption resonance integral, I_a , for ^{238}U over $1 \leq E \leq 10^5$ eV.

- Compute I_a .
- How does this value compare with the value in FNRP Table 3.2?
- What is the corresponding effective microscopic absorption cross section, $\bar{\sigma}_a$, for a $1/E$ flux spectrum over the same energy interval?

Use absorption, including radiative capture and fission, rather than radiative capture alone. Report the resonance integral and effective cross section in barns.

SOLUTION (adapted for Fall 2026):

The resonance integral and corresponding effective cross section are

$$I_a = \int_1^{10^5} \frac{\sigma_a(E)}{E} dE, \quad \bar{\sigma}_a = \frac{I_a}{\int_1^{10^5} E^{-1} dE} = \frac{I_a}{\ln(10^5)}.$$

Using OpenMC pointwise data with $\sigma_a = \sigma_\gamma + \sigma_f$:

```
import openmc
import numpy as np

u238 = openmc.data.IncidentNeutron.from_hdf5(data_path + "U238.h5")
sig_g = u238.reactions[102].xs["294K"]
sig_f = u238.reactions[18].xs["294K"]

E = np.logspace(0, 5, 1000000)
I_a = np.trapz((sig_g(E) + sig_f(E))/E, E)
sig_a_bar = I_a / np.log(1e5)
print(I_a, sig_a_bar)
```

The legacy solution notebook gives

$$I_a \approx 2.75 \times 10^2 \text{ b}, \quad \bar{\sigma}_a \approx 23.9 \text{ b}.$$

The integral is close to the FNRP Table 3.2 value of about 278 b.

Problem 2. A major approximation in Eqs. (3.50)–(3.53) is the explicit assumption that $\phi(E) = 1/E$. Instead, use the same nuclear data as in Problem 1, supplemented by the total cross sections needed for the mixture, with the narrow-resonance (NR) spectrum

$$\phi_{\text{NR}}(E) \propto \frac{1}{\Sigma_t(E)E}.$$

Consider an infinite, homogeneous 100-to-1 atom mixture of ^1H and ^{238}U , where $\Sigma_t(E)$ is the total macroscopic cross section of the mixture.

- Compute the effective microscopic absorption cross section, $\bar{\sigma}_a$, for ^{238}U over $1 \leq E \leq 10^5$ eV, using the NR spectrum as the flux weighting.
- Explain why this value differs from the value computed in Problem 1.

SOLUTION (adapted for Fall 2026):

The flux-weighted effective cross section is

$$\bar{\sigma}_a = \frac{\int_1^{10^5} \sigma_a^{238}(E) \phi_{\text{NR}}(E) dE}{\int_1^{10^5} \phi_{\text{NR}}(E) dE}.$$

Because the ^{238}U number density cancels, the mixture total can be formed from microscopic cross sections as

$$\Sigma_t(E) \propto 100\sigma_t^1\text{H}(E) + \sigma_t^{238}\text{U}(E).$$

The OpenMC-data calculation is

```
h1 = openmc.data.IncidentNeutron.from_hdf5(data_path + "H1.h5")
sig_s_u238 = u238.reactions[2].xs["294K"]
sig_g_h1 = h1.reactions[102].xs["294K"]
sig_s_h1 = h1.reactions[2].xs["294K"]

sig_t_u238 = sig_g(E) + sig_f(E) + sig_s_u238(E)
sig_t_h1 = sig_g_h1(E) + sig_s_h1(E)
phi_nr = 1.0 / ((100*sig_t_h1 + sig_t_u238)*E)
sig_a_bar = np.trapz((sig_g(E) + sig_f(E))*phi_nr, E) / np.trapz(phi_nr, E)
```

The legacy solution notebook gives

$$\bar{\sigma}_a \approx 8.53 \text{ b}.$$

This is much smaller than the $1/E$ result because the NR spectrum includes energy self-shielding. The flux is depressed near ^{238}U absorption resonances, so those large cross sections receive less flux weight than they do under an unshielded $1/E$ spectrum.

13. HOMEWORK 13: LESSON 13

Problem 1. One can compute effective cross sections for a coarse group structure based on the multigroup fluxes and cross sections defined on a finer group structure. Let a fine energy group g be defined over the energies $E_g < E \leq E_{g-1}$. Suppose a coarse group G is defined over $E_{g+1} < E \leq E_{g-1}$ (i.e., over groups g and $g+1$). The coarse-group flux can then be written as

$$\begin{aligned}\phi_G^{\text{coarse}} &= \int_{E_{g+1}}^{E_{g-1}} \phi(E) dE \\ &= \int_{E_g}^{E_{g-1}} \phi(E) dE + \int_{E_{g+1}}^{E_g} \phi(E) dE \\ &= \phi_g^{\text{fine}} + \phi_{g+1}^{\text{fine}}.\end{aligned}$$

- Determine an expression for $\sigma_{a,G}$ in terms of the fine-group fluxes and cross-section values $\sigma_{a,g}$ and $\sigma_{a,g+1}$.
- Assuming the fast and epithermal fluxes are identical, determine the effective σ_a for ^{238}U over $1 < E < 10^7$ eV using FNR Table 3.2.

SOLUTION (adapted for Fall 2026):

The reaction rate per atom integrated over the coarse group must be preserved:

$$\phi_G^{\text{coarse}} \sigma_{a,G} = \phi_g^{\text{fine}} \sigma_{a,g} + \phi_{g+1}^{\text{fine}} \sigma_{a,g+1}.$$

Therefore,

$$\sigma_{a,G} = \frac{\phi_g^{\text{fine}} \sigma_{a,g} + \phi_{g+1}^{\text{fine}} \sigma_{a,g+1}}{\phi_g^{\text{fine}} + \phi_{g+1}^{\text{fine}}}.$$

For part (b), Table 3.2 gives $I_a \approx 278$ b for the epithermal group, so

$$\sigma_a^{\text{epi}} = 0.0869 I_a \approx 24.16 \text{ b}.$$

The fast-group value is $\sigma_a^{\text{fast}} = 0.361$ b. With identical fast and epithermal fluxes,

$$\bar{\sigma}_a = \frac{24.16 + 0.361}{2} \approx 12.26 \text{ b}.$$

Problem 2. Consider an infinite solution of water and uranium with a uniform source of 1 MeV neutrons that emits $100 \text{ n cm}^{-3} \text{ s}^{-1}$. Using the data in Table 1, compute the volumetric absorption and fission rates in the solution.

Table 1. Two-group data for Problems 2 and 3. Macroscopic cross sections are given in cm^{-1} .

	$g = 1$	$g = 2$
Σ_γ	2.04×10^{-3}	1.91×10^{-2}
Σ_f	2.96×10^{-4}	1.38×10^{-2}
$\bar{\nu}\Sigma_f$	7.38×10^{-4}	3.36×10^{-2}
$\Sigma_{s,1 \leftarrow g}$	7.21×10^{-1}	0
$\Sigma_{s,2 \leftarrow g}$	4.27×10^{-2}	2.72×10^0

SOLUTION (adapted for Fall 2026):

The two-group equations for this fixed-source infinite system are

$$\begin{aligned}\Sigma_{r,1}\phi_1 &= \bar{\nu}\Sigma_{f,1}\phi_1 + \bar{\nu}\Sigma_{f,2}\phi_2 + S_1, \\ \Sigma_{a,2}\phi_2 &= \Sigma_{s,2 \leftarrow 1}\phi_1.\end{aligned}$$

Thus,

$$\phi_2 = \frac{\Sigma_{s,2 \leftarrow 1}\phi_1}{\Sigma_{a,2}}, \quad \phi_1 = \frac{S_1}{\Sigma_{r,1} - \bar{\nu}\Sigma_{f,1} - \bar{\nu}\Sigma_{f,2}\Sigma_{s,2 \leftarrow 1}/\Sigma_{a,2}}.$$

The fast removal cross section is

$$\Sigma_{r,1} = 2.04 \times 10^{-3} + 2.96 \times 10^{-4} + 4.27 \times 10^{-2} \approx 0.0450 \text{ cm}^{-1},$$

and

$$\Sigma_{a,2} = 1.91 \times 10^{-2} + 1.38 \times 10^{-2} = 0.0329 \text{ cm}^{-1}.$$

Therefore,

$$\phi_1 \approx \frac{100}{0.0450 - 7.38 \times 10^{-4} - (3.36 \times 10^{-2})(4.27 \times 10^{-2})/0.0329} \approx 1.45 \times 10^5 \text{ cm}^{-2} \text{ s}^{-1},$$

and

$$\phi_2 = \frac{(4.27 \times 10^{-2})(1.45 \times 10^5)}{0.0329} \approx 1.88 \times 10^5 \text{ cm}^{-2} \text{ s}^{-1}.$$

The volumetric absorption rate is

$$\begin{aligned}R_a &= \phi_1 \Sigma_{a,1} + \phi_2 \Sigma_{a,2} \\ &= 1.45 \times 10^5 (2.04 \times 10^{-3} + 2.96 \times 10^{-4}) + 1.88 \times 10^5 (0.0329) \\ &\approx \boxed{6.5 \times 10^3 \text{ s}^{-1} \text{ cm}^{-3}}.\end{aligned}$$

The volumetric fission rate is

$$\begin{aligned}R_f &= \phi_1 \Sigma_{f,1} + \phi_2 \Sigma_{f,2} \\ &= 1.45 \times 10^5 (2.96 \times 10^{-4}) + 1.88 \times 10^5 (1.38 \times 10^{-2}) \\ &\approx \boxed{2.6 \times 10^3 \text{ s}^{-1} \text{ cm}^{-3}}.\end{aligned}$$

Problem 3. Compute k_∞ for the system described in Problem 2.

SOLUTION (adapted for Fall 2026):

For the corresponding homogeneous eigenvalue problem,

$$k_\infty = \frac{\bar{v}\Sigma_{f,1} + \bar{v}\Sigma_{f,2}\Sigma_{s,2\leftarrow 1}/\Sigma_{a,2}}{\Sigma_{r,1}}.$$

Substitution gives

$$k_\infty = \frac{7.38 \times 10^{-4} + (3.36 \times 10^{-2})(4.27 \times 10^{-2})/0.0329}{0.0450} \approx \boxed{0.985}.$$

14. HOMEWORK 14: LESSON 14

Problem 1. Consider a very large graphite block with density 2.26 g cm^{-3} and a uniformly distributed source of 1 MeV neutrons with strength $s_0 \text{ n cm}^{-3} \text{ s}^{-1}$.

Using the data supplied in Table 3.2 of FNRP, construct the necessary cross sections and then compute a three-group approximation of the neutron energy spectrum. State all assumptions.

SOLUTION (adapted for Fall 2026):

Modeling decisions and available data. The problem asks for a coarse spectrum, not a detailed transport calculation, so the first engineering decision is to use the group structure implied by FNRP Table 3.2:

$$\begin{aligned} g = 1 & : 10^5 < E \leq 10^6 \text{ eV}, \\ g = 2 & : 1 < E \leq 10^5 \text{ eV}, \\ g = 3 & : E \leq 1 \text{ eV}. \end{aligned}$$

The 1 MeV source is therefore a group-1 source:

$$S_1 = s_0, \quad S_2 = S_3 = 0.$$

Let $\phi_g = \int_g \phi(E) dE$ be the group-integrated flux. For isotope i , define microscopic data with the isotope index as a superscript:

$$\sigma_{a,g}^{(i)} = \sigma_{\gamma,g}^{(i)} + \sigma_{f,g}^{(i)}, \quad \Sigma_{x,g} = \sum_i N^{(i)} \sigma_{x,g}^{(i)} 10^{-24},$$

where microscopic cross sections are in barns and atom densities are in atoms cm^{-3} .

The infinite-medium group balance is

$$(1) \quad \Sigma_{r,g} \phi_g = S_g + \sum_{\substack{h=1 \\ h \neq g}}^3 \Sigma_{s,g \leftarrow h} \phi_h + \frac{\chi_g}{k_\infty} s_f''.$$

For Problem 1 there is no fission source, so the final term is zero. Removal from a group counts absorption plus scattering out of that group:

$$\Sigma_{r,g} = \Sigma_{a,g} + \sum_{\substack{h=1 \\ h \neq g}}^3 \Sigma_{s,h \leftarrow g}.$$

Prepare the cross-section data. FNRP Table 3.2 gives thermal averages, resonance integrals, and fission-spectrum fast averages. The fast values below are therefore $\chi(E)$ -averaged values, not literal 1 MeV point values. For the intermediate group, use the resonance-integral weighting

$$\bar{\sigma}_{x,2} = \frac{I_x}{\int_1^{10^5} dE/E} = \frac{I_x}{\ln(10^5)}.$$

Thus $\Delta u_2 = \ln(10^5) = 11.512925465$. The table supplies intermediate I_f and I_a ; capture is obtained from $\sigma_{\gamma,2} = \sigma_{a,2} - \sigma_{f,2}$. Since Table 3.2 does not list intermediate scattering averages, use the tabulated thermal scattering average for group 2.

Isotope	Group	σ_f	σ_γ barns	σ_a	σ_s
^{12}C	1	0	1.230×10^{-3}	1.230×10^{-3}	2.36
^{12}C	2	0	1.329×10^{-4}	1.329×10^{-4}	4.81
^{12}C	3	0	3.000×10^{-3}	3.000×10^{-3}	4.81
^{235}U	1	1.220	0.0700	1.290	6.33
^{235}U	2	23.626	11.465	35.091	15.0
^{235}U	3	505.0	86.0	591.0	15.0
^{238}U	1	0.304	0.0570	0.361	7.42
^{238}U	2	1.737×10^{-4}	24.1466	24.1468	9.37
^{238}U	3	1.05×10^{-5}	2.41999	2.420	9.37

The remaining decision is how to map elastic scattering into group-to-group transfer. For isotope i , define

$$\alpha^{(i)} = \left(\frac{A^{(i)} - 1}{A^{(i)} + 1} \right)^2, \quad \xi^{(i)} = 1 + \frac{\alpha^{(i)}}{1 - \alpha^{(i)}} \ln \alpha^{(i)}.$$

For elastic scattering with isotropic center-of-mass scattering, the outgoing energy from incident energy E lies in $\alpha^{(i)}E \leq E' \leq E$. Therefore upscatter terms vanish in this slowing-down model. A single elastic collision also cannot carry a group-1 neutron directly to group 3 for these nuclides, so the only off-diagonal transfer terms are $2 \leftarrow 1$ and $3 \leftarrow 2$. With a $1/E$ spectrum inside each slowing-down group,

$$\sigma_{s,g+1 \leftarrow g}^{(i)} \simeq \sigma_{s,g}^{(i)} \frac{\xi^{(i)}}{\Delta u_g}, \quad \Delta u_g = \ln \left(\frac{E_{g-1}}{E_g} \right).$$

Here $\Delta u_1 = \ln(10)$ and $\Delta u_2 = \ln(10^5)$.

Isotope	$\alpha^{(i)}$	$\xi^{(i)}$	$\sigma_{s,2 \leftarrow 1}^{(i)}$ barns	$\sigma_{s,3 \leftarrow 2}^{(i)}$
^{12}C	0.7159763314	0.1577689898	0.161703	0.0659145
^{235}U	0.9831226659	0.0084865458	0.0233302	0.0110570
^{238}U	0.9833336251	0.0083798718	0.0270038	0.0068201

Assemble and solve the graphite balance. For pure graphite,

$$N^{(C)} = \frac{\rho_C}{M_C} N_A = \frac{2.26}{12.0} (6.02214076 \times 10^{23}) = 1.13416984 \times 10^{23} \text{ atoms cm}^{-3}.$$

Thus one barn corresponds to $0.113416984 \text{ cm}^{-1}$ for this graphite. The nonzero macroscopic cross sections are

$$\begin{aligned} \Sigma_{a,1} &= 1.39503 \times 10^{-4}, & \Sigma_{a,2} &= 1.50724 \times 10^{-5}, & \Sigma_{a,3} &= 3.40251 \times 10^{-4}, \\ \Sigma_{s,2 \leftarrow 1} &= 1.83399 \times 10^{-2}, & \Sigma_{s,3 \leftarrow 2} &= 7.47582 \times 10^{-3}. \end{aligned}$$

All values are in cm^{-1} . The fixed-source balances reduce to

$$\begin{aligned} (2) \quad & (\Sigma_{a,1} + \Sigma_{s,2 \leftarrow 1}) \phi_1 = s_0, \\ (3) \quad & (\Sigma_{a,2} + \Sigma_{s,3 \leftarrow 2}) \phi_2 = \Sigma_{s,2 \leftarrow 1} \phi_1, \\ (4) \quad & \Sigma_{a,3} \phi_3 = \Sigma_{s,3 \leftarrow 2} \phi_2. \end{aligned}$$

Solving sequentially gives

$$\begin{aligned} \phi_1 &= \frac{s_0}{\Sigma_{a,1} + \Sigma_{s,2 \leftarrow 1}} = 54.1144 s_0, \\ \phi_2 &= \frac{\Sigma_{s,2 \leftarrow 1} \phi_1}{\Sigma_{a,2} + \Sigma_{s,3 \leftarrow 2}} = 132.488 s_0, \end{aligned}$$

and

$$\phi_3 = \frac{\Sigma_{s,3 \leftarrow 2} \phi_2}{\Sigma_{a,3}} = 2910.95 s_0.$$

Since s_0 has units $\text{n cm}^{-3} \text{ s}^{-1}$, these coefficients have units of cm and the fluxes have units $\text{n cm}^{-2} \text{ s}^{-1}$.

Interpretation and checks. The balance check is

$$\Sigma_{a,1}\phi_1 + \Sigma_{a,2}\phi_2 + \Sigma_{a,3}\phi_3 = s_0$$

to rounding. The result is physically consistent with a nonmultiplying graphite moderator: most source neutrons scatter down in energy and are eventually absorbed after reaching the thermal group.

Problem 2. Now consider the case in which uranium enriched to 2% ^{235}U by atom is uniformly dispersed in the graphite block. Determine the resulting infinite-medium multiplication factor k_∞ as a function of the C-to-U atomic ratio $R = N_C/N_U$. State all assumptions.

SOLUTION (adapted for Fall 2026):

Keep the same model, but change the material. Use the same group structure, microscopic data, and scattering-transfer model as in Problem 1. The material composition is now

$$N_U = \frac{N_C}{R}, \quad N^{(235)} = 0.02 \frac{N_C}{R}, \quad N^{(238)} = 0.98 \frac{N_C}{R}.$$

Let

$$c = N_C 10^{-24} = 0.113416984.$$

For any microscopic reaction coefficient in uranium, define the enrichment average

$$u_{x,g} = 0.02 \sigma_{x,g}^{(235)} + 0.98 \sigma_{x,g}^{(238)}.$$

From the prepared data in Problem 1,

g	$u_{f,g}$	$u_{a,g}$	$u_{\gamma,g}$
1	0.32232	0.37958	0.05726
2	0.47268264	24.36565761	23.89297497
3	10.10001029	14.1916	4.09158971

in barns. The corresponding uranium-mixture transfer coefficients are

$$u_{s,2 \leftarrow 1} = 0.02693037, \quad u_{s,3 \leftarrow 2} = 0.00690485$$

in barns.

Build the R -dependent macroscopic coefficients. The absorption and fission coefficients are

$$\Sigma_{a,g}(R) = c \left(\sigma_{a,g}^{(C)} + \frac{u_{a,g}}{R} \right), \quad \Sigma_{f,g}(R) = c \frac{u_{f,g}}{R}.$$

The off-diagonal scattering transfers are

$$\Sigma_{s,2 \leftarrow 1}(R) = c \left(0.161702956 + \frac{u_{s,2 \leftarrow 1}}{R} \right),$$

$$\Sigma_{s,3 \leftarrow 2}(R) = c \left(0.065914510 + \frac{u_{s,3 \leftarrow 2}}{R} \right).$$

Solve the homogeneous eigenvalue problem. For an infinite homogeneous multiplying medium, the source-free group balances are

$$(5) \quad (\Sigma_{a,1} + \Sigma_{s,2 \leftarrow 1}) \phi_1 = \frac{s_f'''}{k_\infty},$$

$$(6) \quad (\Sigma_{a,2} + \Sigma_{s,3 \leftarrow 2}) \phi_2 = \Sigma_{s,2 \leftarrow 1} \phi_1,$$

$$(7) \quad \Sigma_{a,3} \phi_3 = \Sigma_{s,3 \leftarrow 2} \phi_2,$$

where

$$s_f''' = \sum_{g=1}^3 \nu \Sigma_{f,g}(R) \phi_g.$$

FNRP Table 3.2 does not give $\bar{\nu}$ values, so the expression below uses a common named value ν . If group- or isotope-dependent yields are used, replace $\nu \Sigma_{f,g}$ by

$$\sum_i N^{(i)} \bar{\nu}_g^{(i)} \sigma_{f,g}^{(i)} 10^{-24}.$$

Normalize the eigenvector by ϕ_1 :

$$x(R) = \frac{\phi_2}{\phi_1}, \quad y(R) = \frac{\phi_3}{\phi_1}.$$

The lower two rows of Eq. (7) give

$$x(R) = \frac{\Sigma_{s,2\leftarrow 1}(R)}{\Sigma_{a,2}(R) + \Sigma_{s,3\leftarrow 2}(R)}, \quad y(R) = \frac{\Sigma_{s,3\leftarrow 2}(R)x(R)}{\Sigma_{a,3}(R)}.$$

Substitution in the fission production-to-absorption balance gives

$$k_{\infty}(R) = \frac{\nu\Sigma_{f,1}(R) + \nu\Sigma_{f,2}(R)x(R) + \nu\Sigma_{f,3}(R)y(R)}{\Sigma_{a,1}(R) + \Sigma_{a,2}(R)x(R) + \Sigma_{a,3}(R)y(R)}.$$

Assumptions to attach to this result. Fast-group cross sections are fission-spectrum averages rather than pointwise 1 MeV values; intermediate absorption and fission values are resonance integrals divided by $\ln(10^5)$; intermediate scattering uses the thermal scattering averages; elastic transfer is treated as downscatter with the $1/E$ weighting above; resonance self-shielding is not included; and a numerical value requires a stated choice of fission neutron yields.

Problem 3. Use OpenMC to compute k_∞ for $R = 1000$ for Problem 2 by adapting our in-class example.

SOLUTION (adapted for Fall 2026):

Translate the analytic model into an OpenMC input. For $R = 1000$, the atom densities implied by the graphite density are

$$\begin{aligned} N_C &= 1.13416984 \times 10^{23} \text{ cm}^{-3}, & N_U &= 1.13416984 \times 10^{20} \text{ cm}^{-3}, \\ N^{(235)} &= 2.26834 \times 10^{18} \text{ cm}^{-3}, & N^{(238)} &= 1.11149 \times 10^{20} \text{ cm}^{-3}. \end{aligned}$$

The OpenMC calculation should therefore use one infinite homogeneous material containing graphite and 2 atom-percent enriched uranium with $N_C/N_U = 1000$. Use reflective or periodic boundary conditions on a simple cell, run a k -eigenvalue calculation, and report the mean and uncertainty after an adequate number of inactive and active batches.

Analytic check value. The three-group model above is not a substitute for OpenMC, but it gives a useful consistency check. If the same three-group closure is evaluated with a common fission neutron yield $\nu = 2.43$ for all fissions, then at $R = 1000$

$$x = 1.78865, \quad y = 6.85861,$$

and

$$k_\infty(R = 1000; \nu = 2.43) = 1.04793.$$

The OpenMC result should be compared against this number as a modeling check, not treated as a continuous-energy benchmark target.

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